

Beyond Open Models: Building Open Search Infrastructure for AI Agents

Open Data, Scalable Retrieval, and Observable Interaction

Prof. Dr. Michael Granitzer

Chair of Data Science, University of Passau

IT:U Interdisciplinary Transformation University Austria

Coordinator OpenWebSearch.EU

CLEF 2026 · Keynote · 22 September 2026



Funded by



Bundesministerium
für Forschung, Technologie
und Raumfahrt

Bayerisches Staatsministerium für
Wirtschaft, Landesentwicklung und Energie



Search is dead — long live search

Hey, Michael. Ready to dive in?

Ask anything

+

Tools

🔊

⬆

Semrush clickstream analysis, Apr 2026 · Chatterji et al., How People Use ChatGPT, NBER 2025 · Chen et al., BrowseComp-Plus, 2025 (GPT-5, o3).

Search is dead — long live search

Hey, Michael. Ready to dive in?



"Search the web" sits behind the box, switched on by default in most assistants.

Semrush clickstream analysis, Apr 2026 · Chatterji et al., How People Use ChatGPT, NBER 2025 · Chen et al., BrowseComp-Plus, 2025 (GPT-5, o3).

Search is dead — long live search

Hey, Michael. Ready to dive in?



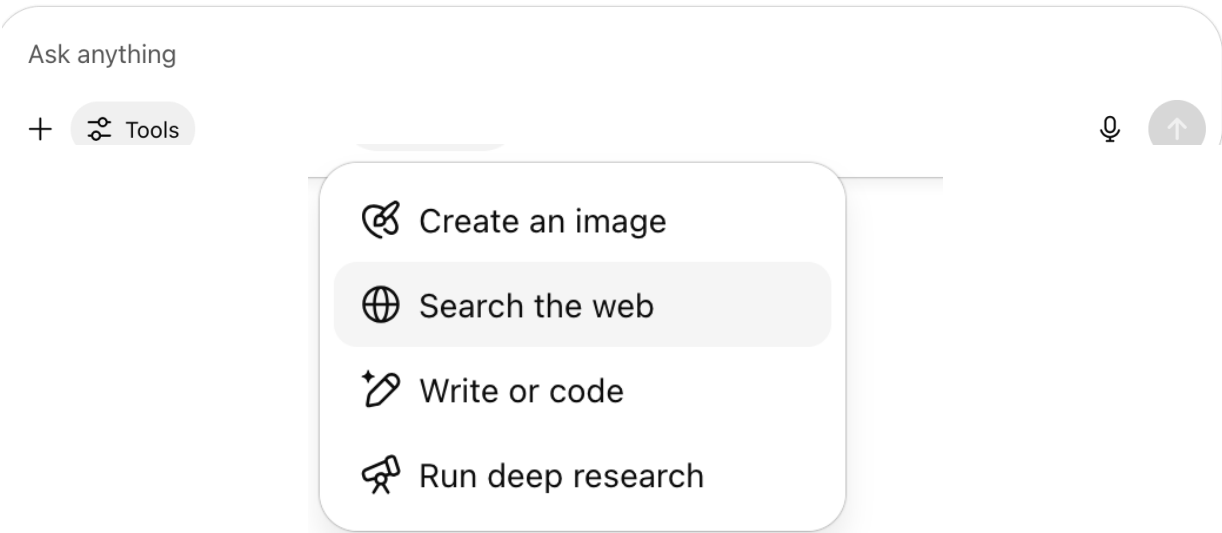
"Search the web" sits behind the box, switched on by default in most assistants.



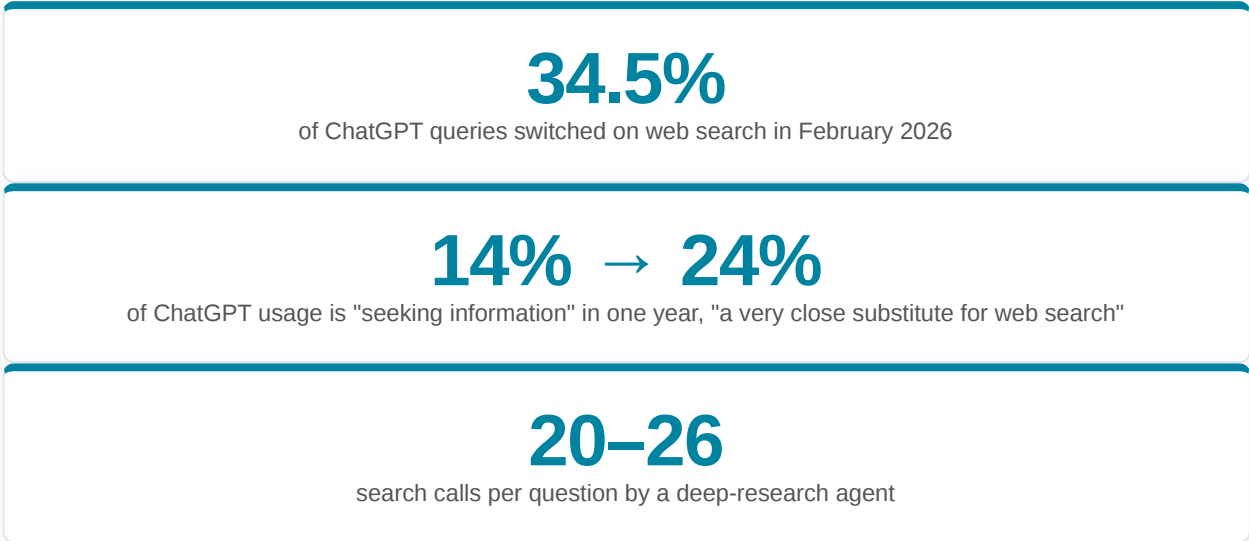
Semrush clickstream analysis, Apr 2026 · Chatterji et al., How People Use ChatGPT, NBER 2025 · Chen et al., BrowseComp-Plus, 2025 (GPT-5, o3).

Search is dead — long live search

Hey, Michael. Ready to dive in?



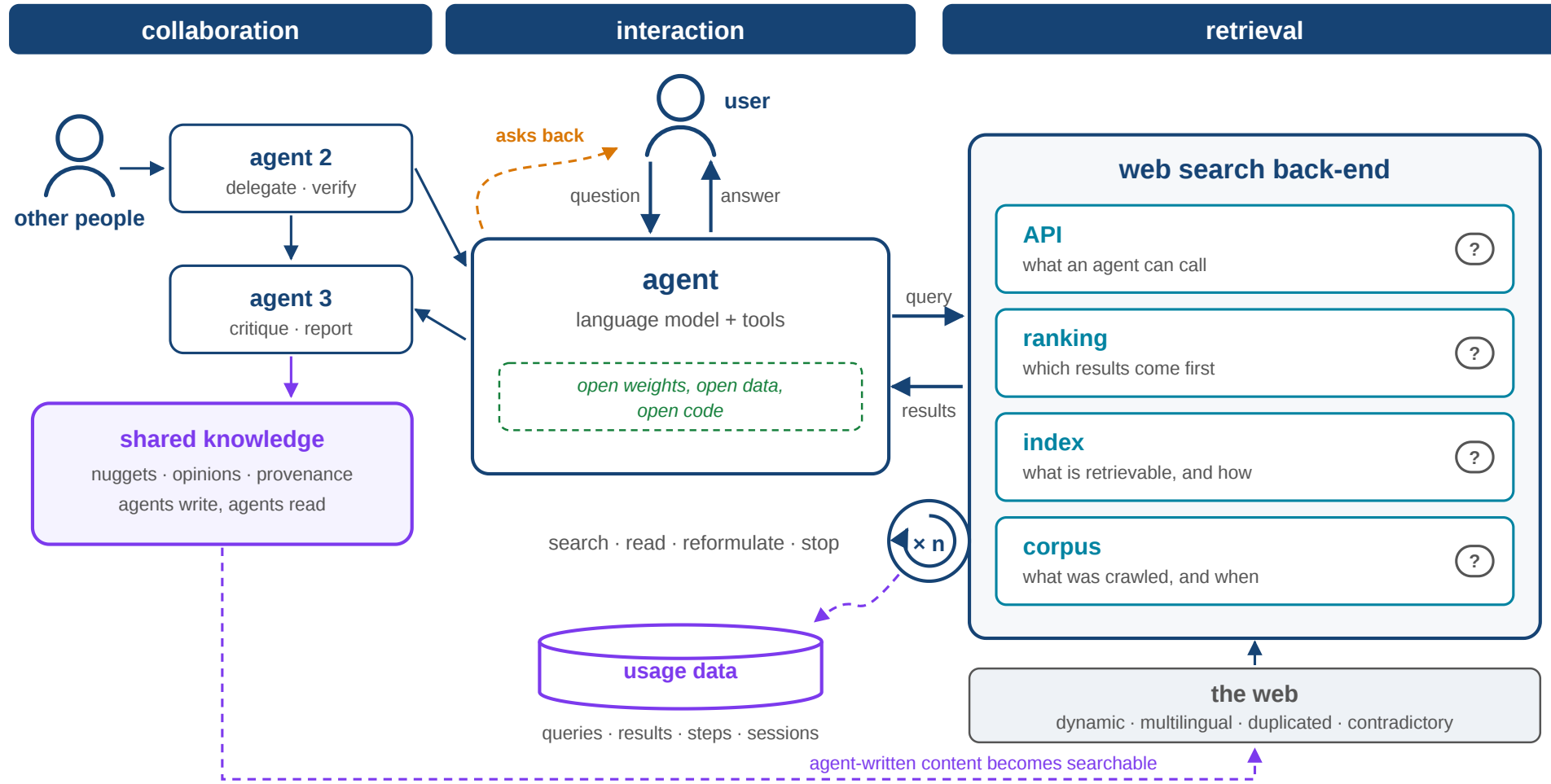
"Search the web" sits behind the box, switched on by default in most assistants.



The box where people type changed. What answers them is still a search engine, queried more often, by something that is not a person.

Semrush clickstream analysis, Apr 2026 · Chatterji et al., How People Use ChatGPT, NBER 2025 · Chen et al., BrowseComp-Plus, 2025 (GPT-5, o3).

Agentic (search) (eco)systems are complex



And this is only the search part. Beyond it the system grows again: tools and MCP, skills and skill learning, workflows and orchestration.

Understanding search meant operating search

What we could observe about users was always bounded by what we operated.

A released web log records two kinds of event

AOL 2006, RELEASED AND WITHDRAWN IN THREE DAYS

A query, and a click on a result. 36.4M queries, 657,426 users, 19.4M clicks — and the only results in the file are the ones that were clicked. Sessions are not marked. ORCAS in 2020 is thinner still: clicks alone, no users, no sessions.

A system you run records fifty-eight

SOWIPOINT, OPERATED BY GESIS, 2014–15

Typing a query, selecting a facet, visiting a document, following a citation, exporting a record. One year gave **558,008 sessions** and about **8M interactions**, of which 179,796 were queries — and the result lists are stored as they were shown.

We did learn from query logs, and obtaining them was the hard part. But a query and a click are the artefacts a search session leaves behind, not the session itself. The richer record came from systems our own community ran.

Reimer et al., The Archive Query Log, SIGIR 2023, Table 1 (AOL and ORCAS figures) · Mayr, Sowipoint User Search Sessions Data Set, 2016, doi:10.7802/1380 · Joachims, KDD 2002 · Joachims, Swaminathan, Schnabel, WSDM 2017.

One hypothesis, four topics

HYPOTHESIS

We have to build full, usable agentic search systems at scale in order to understand them.

Components can be studied on static collections. Systems cannot: what they do depends on the corpus, on the traffic, and on the people and agents using them.

TOPIC 1

Building a sovereign search system

OpenWebSearch.EU and the Open Web Index: web data, an index and a search API that a research community owns and operates.

TOPIC 2

Scale and robustness as research variables

What changes when the corpus grows and the text gets messy, and the instruments that make both measurable.

TOPIC 3

Usage data: simulate and pool

We missed much of the interaction data of web search. We should not miss the interaction data of agentic search.

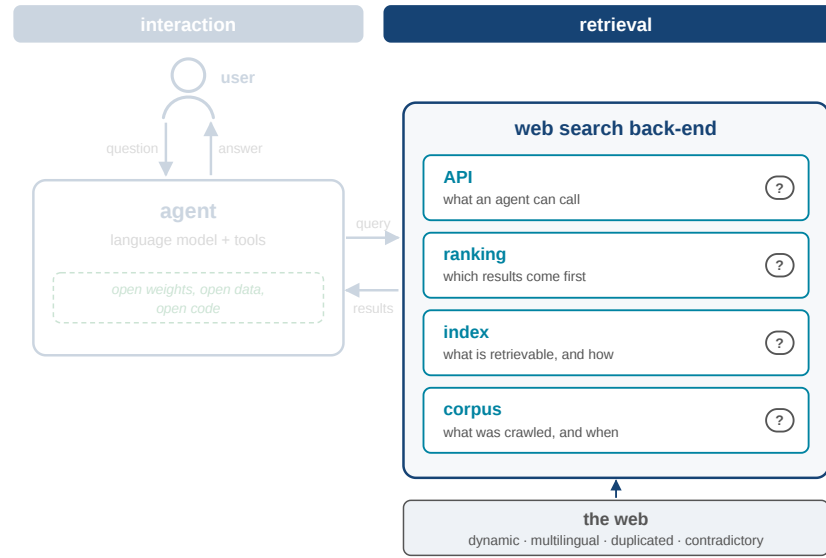
TOPIC 4

Societies of humans and models

Beyond one user and one system: many agents, many people, and the knowledge they share.

Topic 1 · Building a sovereign search system

OPENWEBSEARCH.EU, THE OPEN WEB INDEX, AND A SEARCH API ON TOP



Looking at the Model-side first

Open weights

Llama, Qwen, DeepSeek, Mistral and gpt-oss can be run and fine-tuned by anyone. Their training data and recipes stay private, so they can be used but not inspected.

Fully open

OLMo 3, Apertus, Soofi S, OpenEuroLLM Pleias, Marin and Comma release weights, data, code and checkpoints.

Open-source models on top of closed retrieval infrastructure remain closed research instruments.

OSI, Open Source AI Definition 1.0, Oct 2024 · AI2, OLMo 3, Nov 2025 · Swiss AI, Apertus, Sep 2025 · SOOFI, Soofi S, Jul 2026.

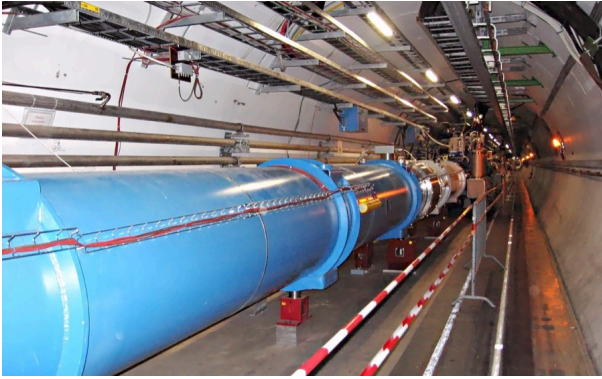
Today's API-based back-ends are not open

Service	Still available	Corpus known	Ranking replaceable	Snapshot fixable	Logs for research
Google Custom Search API	ends Jan 2027	✗	✗	✗	✗
Bing Web Search	retired Aug 2025	✗	✗	✗	✗
Brave Search API	✓	claims 40B pages	re-rank only	✗	✗
Exa · Tavily · Perplexity	✓	✗	✗	✗	✗
OpenAI · Anthropic search tools	✓	✗	✗	✗	✗

So what remains is that you have to trust the web search engine to return the right results and account for its uncertainty.

Research sovereignty

Methodological autonomy: the ability to inspect, reproduce, vary, operate and evaluate the whole information-access process.



CERN · particle physics.

25 member states, one accelerator.



ITER · fusion.

35 nations, one reactor none could build alone.



Webb · astrophysics.

Three agencies fly it, the observations go to a public archive.



Copernicus · climate.

Earth observation, free and open by policy.



the web index, and the ranking



the social record



Hugging Face

the models and the datasets



the code · Microsoft since 2018

Our field's instruments are privately owned. No costs, no control. What could go wrong?

Photographs: J. Herzog CC BY-SA 3.0 · ITER Organization CC BY 2.0 · NASA/MSFC/D. Higginbotham · Rama CC BY-SA 2.0 fr. Wordmarks belong to their owners.

What could go wrong

Every one of these was, at the time, infrastructure that somebody's research depended on.



2018

Microsoft buys GitHub for
7.5 bn USD



2023

Twitter ends free API
access; Reddit prices its
own



2024

Meta shuts down
CrowdTangle



2025

Bing Search APIs retired



2026

NVIDIA agrees to acquire
Hugging Face for
12.9 bn USD

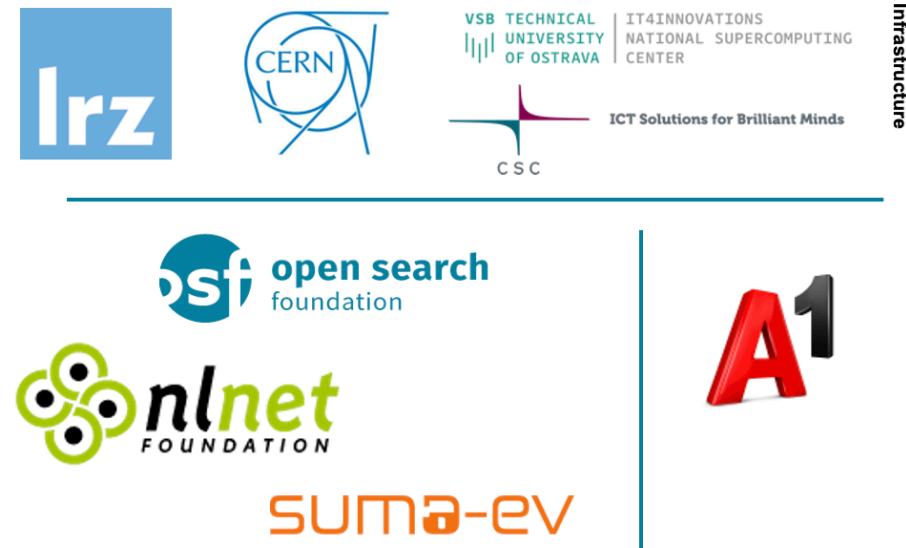
Openness of an artefact is not enough. The infrastructure it lives on has to be a jointly managed public good: CERN for physics, and through CERN, Zenodo for everyone's data; EuroHPC for compute. **For the web there is no equivalent.**

Microsoft newsroom, 4 Jun 2018 · Murtfeldt et al., RIP Twitter API, 2024 · Microsoft lifecycle notice, May 2025 · NVIDIA blog and Form 8-K, 2–3 Sep 2026 · Zenodo, operated by CERN with OpenAIRE and EU funding.

OpenWebSearch.EU: building the index together



Research



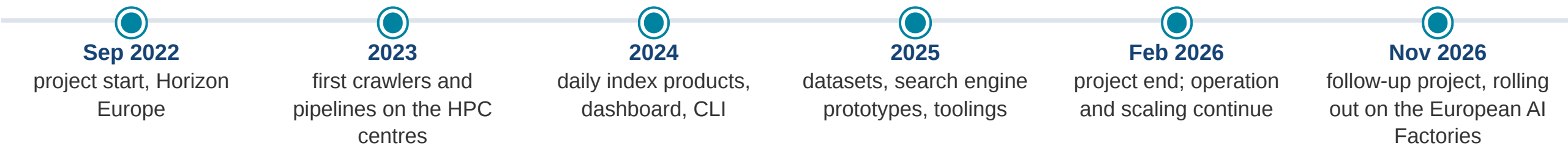
NGOs



Businesses

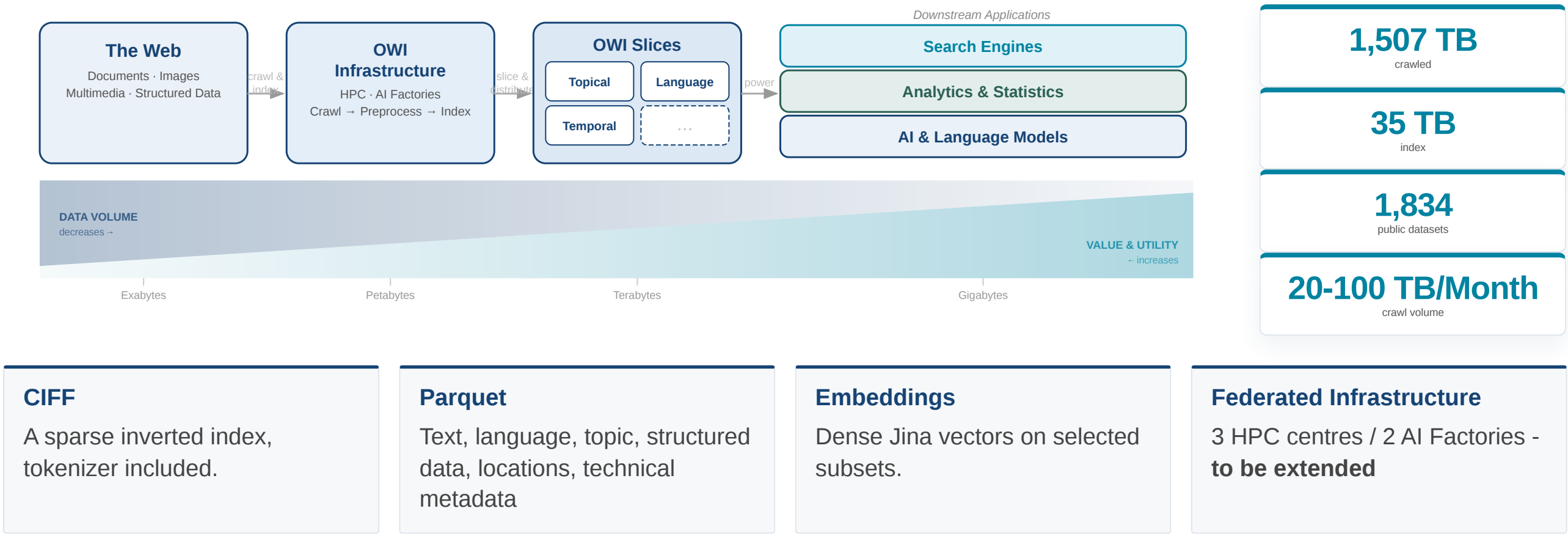
Objectives

- 1. Build an open index of the Web
- 2. Build a corresponding web search infrastructure
- 3. Establish a network of infrastructure partners.
- 4. Bootstrap an ecosystem



Granitzer et al., Impact and development of an Open Web Index for open web search, JASIST 75(5), 2024.

The Open Web Index: web data every day, ready to use



Dashboard figures May 2026 (openwebindex.eu).

OpenWebIndex.eu: Dashboard and Access

Open Web Index

Search this site

⌕ K

→] Sign in

▼

Overview

EXPLORE

OWI Statistics

Datasets

Corpora

OWI Lakes

Models

CONTROL

Site Console

Request removal

Your data

RESOURCES

Crawler

Workflows

Repositories

Lists

ABOUT

News

Get started

Partners

Team

Applications

index.eu/datasets

The Open Web Index

Your daily dose of web data

The Open Web Index is a public index of the web, built so that search and analysis are not the private property of a handful of companies. This dashboard is how you look into it: what it holds, who crawled it, where it is stored, and what you may do with it.

Where to start

Read the paper behind it

COLLECTIONS

955

mirrored here

DOCUMENTS

14B

across collections

VOLUME

57.6 TB

uncompressed

DATACENTERS

3

contributing crawl data

Figures cover the slice of the index this deployment mirrors — [see the full statistics](#).

Where to go

The Open Web Index is a public index of the web, built by the OpenWebSearch.eu consortium so that search and analysis are not the private property of a handful of companies. This dashboard is how you look into it: what it contains, where it is stored, what you may do with it, and how to get at the data yourself.

See what the index holds

Statistics show crawl volume, dataset counts and language mix over time, and every figure states which slice of the index it covers. A deployment may mirror only part of the index — a number here is never a claim about the whole web.

Browse datasets and corpora

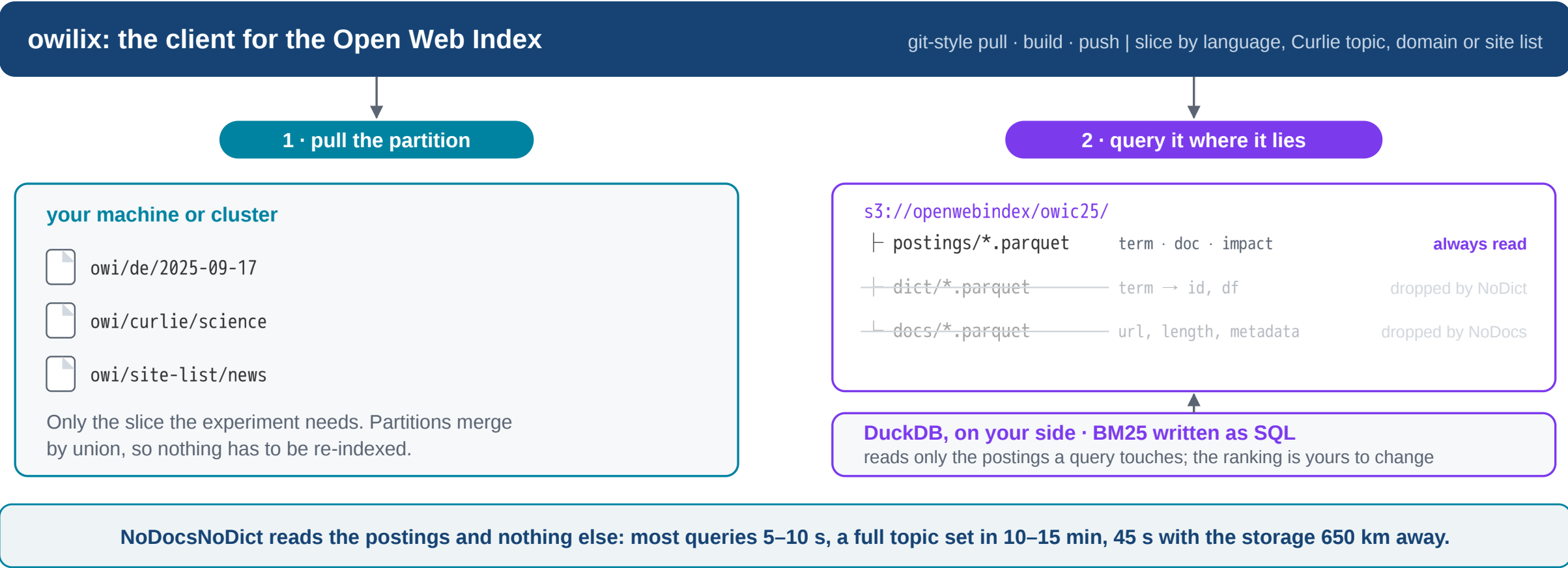
The catalog lists the collections mirrored into our object store, with their DataCite metadata, file listings and previews. Downloads are served straight from object storage rather than proxied through this site.

TODO: screenshot plus slides

≡

13

Two ways in: pull the index, or query it where it lies



ECIR 2026 · HENDRIKSEN, HIEMSTRA, DE VRIES

Open Web Indexes for Remote Querying

- QUESTION

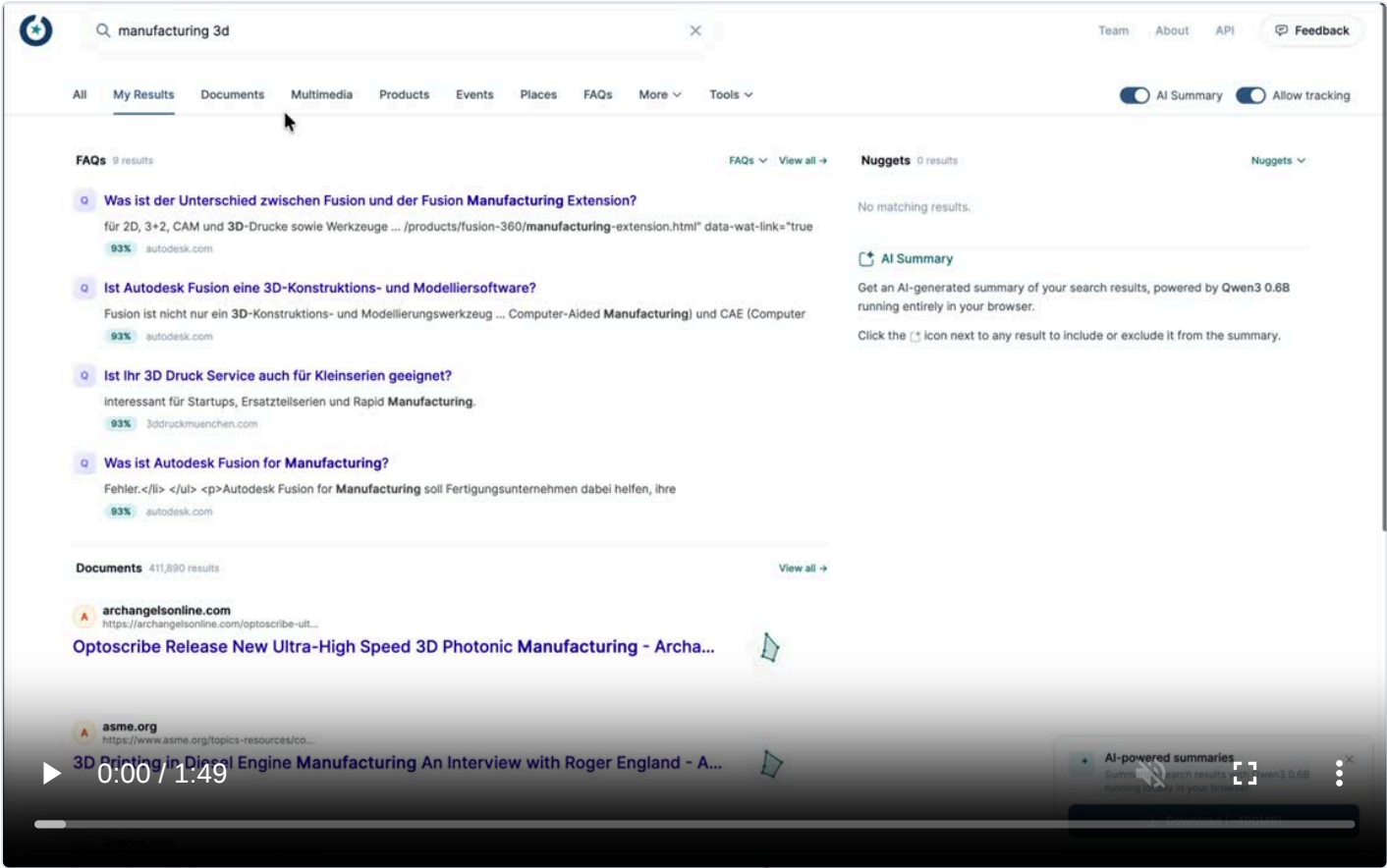
Can a web-scale index be shared as plain files and queried where it lies, instead of behind an API?
- SETUP

ClueWeb09-B, 12-B and 22-B as compressed Parquet on S3; BM25 run client-side in DuckDB; four schemas, each dropping one join.
- CONCLUSION

Seconds, not milliseconds. Nobody has to host a copy, and the retrieval model, the snapshot and the sub-collection are yours.



OURRS.eu: the open web search API plus MCP



ourrs.eu, index snapshot September 2026; screencast September 2026.

1 Billion

pages

950M

typed entities

15

verticals, plus an API and MCP

4

isolated corpora

Persisted AI Summaries

Agentic Relevance Feedback

The Open Web Index Research Retrieval System - OURRS: an early web search prototype over a billion pages of the index (September 2026)

Where the OWI goes next

Scaling Up and Out

- **Quantity:** Crawling across European Data Centers and AI Factories at scale
 - New project for operationalising the OWI will start November 2026
- **Quality:** Headless rendering, GenAI excluding
- Multimedia and multimodal data
- **Tooling:** improved cli tooling, potential access to raw data, corpora & model management

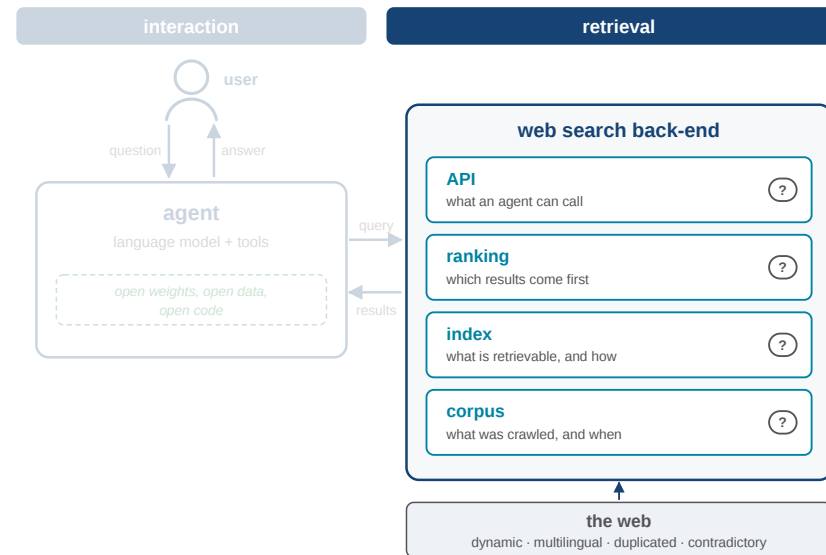
OWI Lakes

- Topical coherent, deduplicated and structured sub-indices
- Every index as Parquet Data Lake on object storage, queried with SQL — remote representations of prebuilt indexes rather than copies to host
- Community-driven, with OWI being the backbone

Hendriksen, Hiemstra, Lin, de Vries, Geese Belong in (Data) Lakes: An Index Lake for Anserini, CIKM 2026.

What the Open Web Index contributes to evaluation

WEB DATA COLLECTION AT SCALE, AND WHAT A LAB CAN DO WITH IT



Today's corpora, and what an open index adds

Collection	Documents	Crawled	What the text is
MS MARCO passage	8.8M passages	2016–2018	cleaned passages
ClueWeb22	10B pages, 207 languages	once, in 2022	raw pages with DOM annotations, under licence
MS MARCO Web Search	a 100M-document benchmark on ClueWeb22	once, in 2022	the same pages, with click labels
LongEval web	dated snapshots, 2022–2023	from a search-engine partner	a partner's documents
BrowseComp-Plus	100,195 documents	fixed, 2025	curated so agents can be compared
Open Web Index	39B URLs crawled, 14B indexed, 57.6 TiB index size	every day	crawled pages, 185 languages, structured data extracted

The OWI adds freshness, because crawl and index run daily; scale, because it is web-scale and growing; richness, as we also provide structured data and see it as community effort, so external contributions and requests are welcomed.



Datasets from the index: language, and place

Untuk menyewa kereta di Girona, anda mestilah berusia sekurang-kurangnya 21 tahun [...] (msa)

How old do I have to be to rent a car in Girona?

In the month March, the average car rental length in Girona is 10 days.

SIGIR 2026 RESOURCE

WebFAQ 2.0 · multilinguality

BUILT FROM

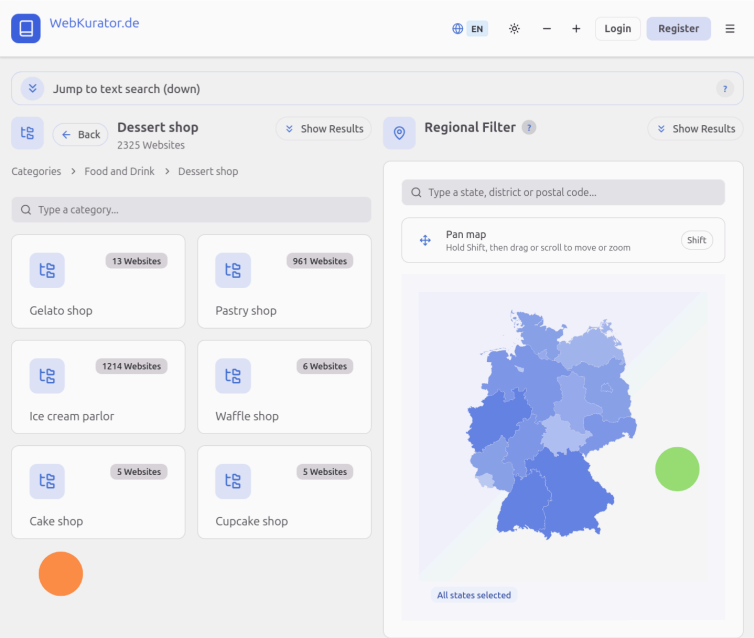
schema.org FAQPage markup, mined and crawled directly. The pairs are written by site owners, not generated.

SIZE

198M question–answer pairs in 108 languages, 14.3M of them aligned across 3,970 language combinations.

CONTROLS

which language, or which language pair, an experiment runs in. 1.25M queries ship with 200 hard negatives each.



WEBSCI 2026

WebKurator · region and topic

BUILT FROM

the legal imprint of 5.54M German-language sites, geocoded and then topic-labelled automatically.

SIZE

2.58M sites in Germany carry both a location and a topic at launch.

CONTROLS

region and topic as two independent axes, so "this topic in that region" is a query, not a filter over one hierarchy.

OWS-Curlie-2025: the collection behind WOWS 2026

The collection

- Six months of our own crawl, March to August 2025, English pages of Curlie-listed sites.
- **20.7M documents**, about 70 GB, with embeddings for the judged parts.
- Three judged splits of 60 to 90k documents. One ships its qrels; two keep them withheld.
- owilix to pull it, ir-datasets-owi to use it.

The shared task

- **WOWS 2026**, the third Workshop on Open Web Search, at ECIR in Delft.
- Topics and judgements written by students in IR courses at Radboud, Jena and Kassel.
- Run with **Webis**: ChatNoir serves it, PyTerrier baselines ship with it, runs go through TIRA.
- The withheld qrels are what turn a dataset into a task.

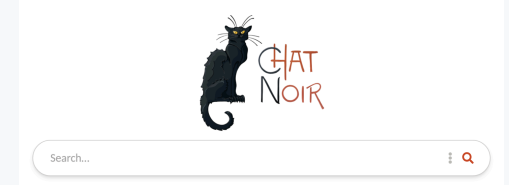
A collection cut from our own crawl, judged in teaching, and evaluated on community infrastructure. No search-engine partner had to agree to any of it.

OpenWebSearch/wows-code, directory ecir26 · ir-datasets-owi · openwebindex.eu/corpora.

Evaluation through an API a partner runs

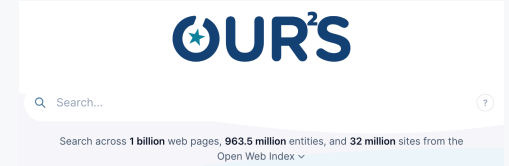
ChatNoir, run by Webis

Frozen collections behind one research API: ClueWeb09, 12 and 22, MS MARCO, LongEval, and OWS-Curлие-2025 as its default index. Static corpora are what make a run citable years later. chatnoir.eu/docs/api-general



OURRS, run in Passau

The same idea for a corpus that changes daily, with agents as first-class clients. Searches **1 billion pages**, **963.5M entities** and **32M sites** of the Open Web Index, plus owic25, Wikipedia and MS MARCO. Eighteen verticals, six ranking profiles, MCP as a two-way channel for feedback.



Works today

- query a live corpus instead of hosting one
- different search verticals: products, places, scholarly, news..
- agents return feedback, summaries and opinions through MCP

In the making

- qrels carry no time stamp, so runs a week apart are not comparable
- six ranking profiles, effectively fixed: no bring-your-own ranker
- no clean export of a corpus slice with its judgements

Summary so far: infrastructure, not only models

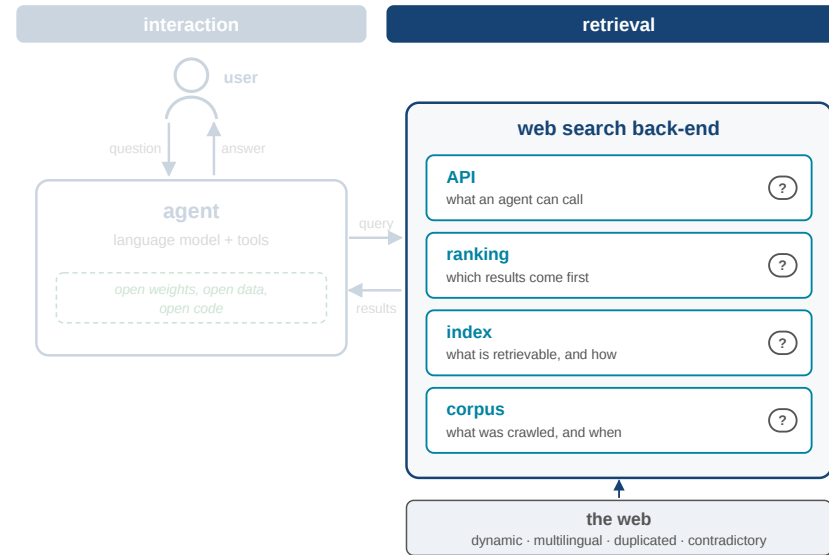
H1 · HYPOTHESIS 1

Openness must extend beyond the model. Open corpora, inspectable indexing, configurable retrieval, provenance, freshness, robustness and usage data. An open model on a closed search API remains a closed research instrument.

- Evaluating an agent-search systems fully means running web-scale retrieval which requires large web-crawls and significant infrastructure.
- No single institute affords that alone. On shared public infrastructure, many institutes can, which is the model of other infrastructure-heavy research fields with high impact
- Building and operating an index and a search engine has become feasible for small teams, in part because coding agents now carry much of the engineering.

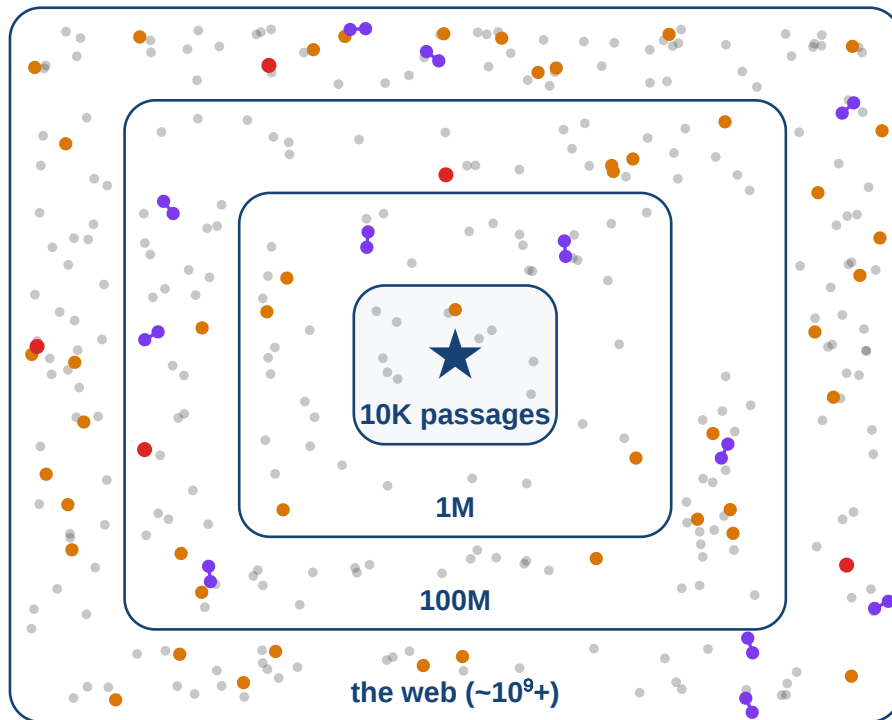
Topic 2 · Scale and robustness as research variables

WHAT CHANGES WHEN THE CORPUS GROWS, AND WHEN THE TEXT IS REAL WEB TEXT



Does scale matter, if we can isolate the effect in a test collection?

same query, same relevant document, a growing corpus around it
the web ($\sim 10^9+$ passages)



always there

- ★ the relevant document
- irrelevant passages

what the ranker must reject grows

- look relevant, but are not
- near-duplicates
- other languages, contradictions
- the cost of holding it all

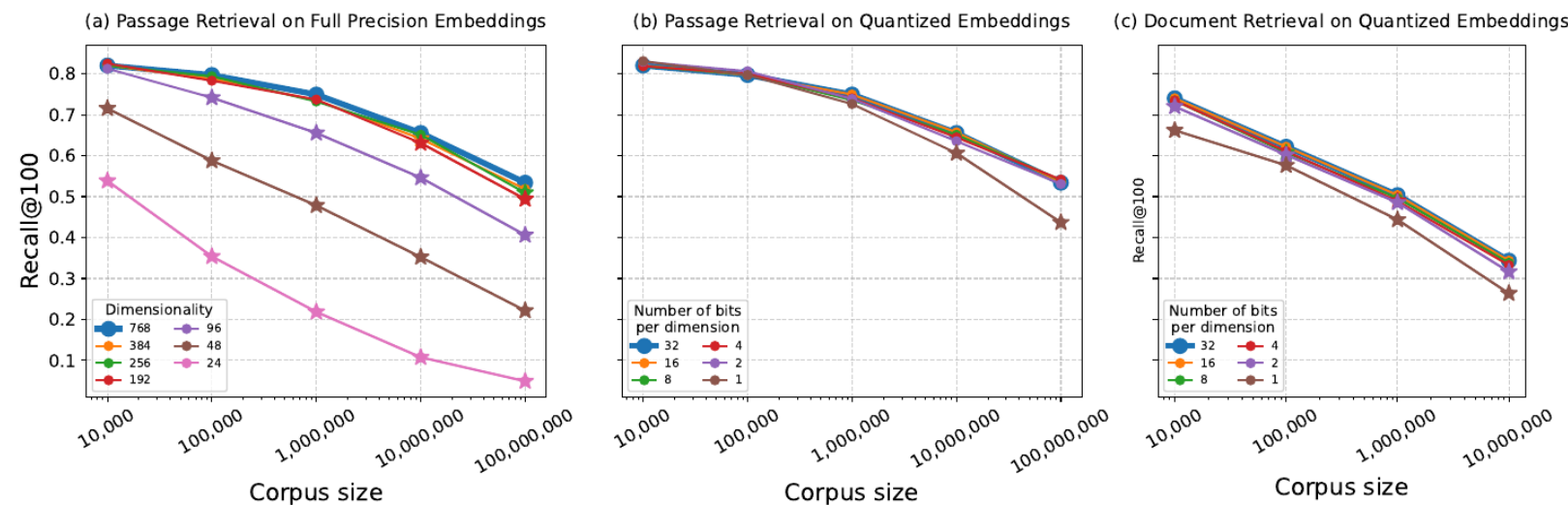
H2 · HYPOTHESIS 2

Scale is a scientific variable.

Corpus size and complexity change effectiveness, efficiency and robustness. A result on a small collection does not carry over to the web by itself.

The relevant document stays one.
What grows is what the ranker has to reject.

CoRECT: A Framework for Evaluating Embedding Compression Techniques at Scale



Caspari et al., CoRECT, Fig. 2. Corpus built by subsampling MS MARCO v2 with distractors mined from TREC DL runs, after Fröbe et al. (ECIR 2025), not by random sampling.

- **Scale alone** costs 29 points of Recall@100 from 10K to 100M, and flips which model is best.
- **Quantization is free at every size**, down to 2 bits per dimension.
- **Truncation and 1-bit are not:** free at 10K, costly at 100M.

ECIR 2026 CoRECT

QUESTION

Do compression methods hold up as the corpus grows?

SETUP

CoRE: the same 65 TREC DL 2023 queries and judgements throughout, the corpus grown from 10K to 100M passages. Each query keeps 10 relevant passages and **100 distractors** mined by the subsampling method of Fröbe et al.; the rest is random filler. Four embedding models, eight compression families.

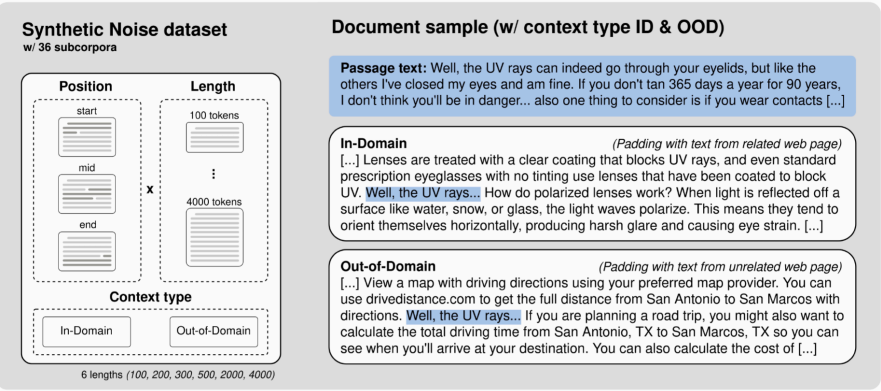
RESULT

The cost of compression grows with the corpus, and the ranking of models flips along the way.

CONCLUSION

A conclusion drawn at 10K passages can be wrong at 100M. Compression must be evaluated at scale.

Robustness: benchmark text is clean, the web is not



SETUP

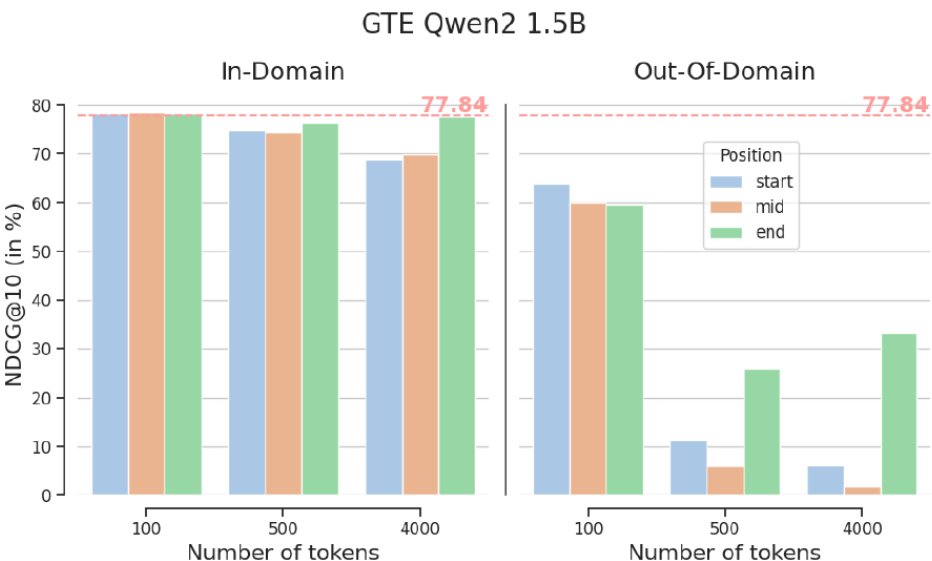
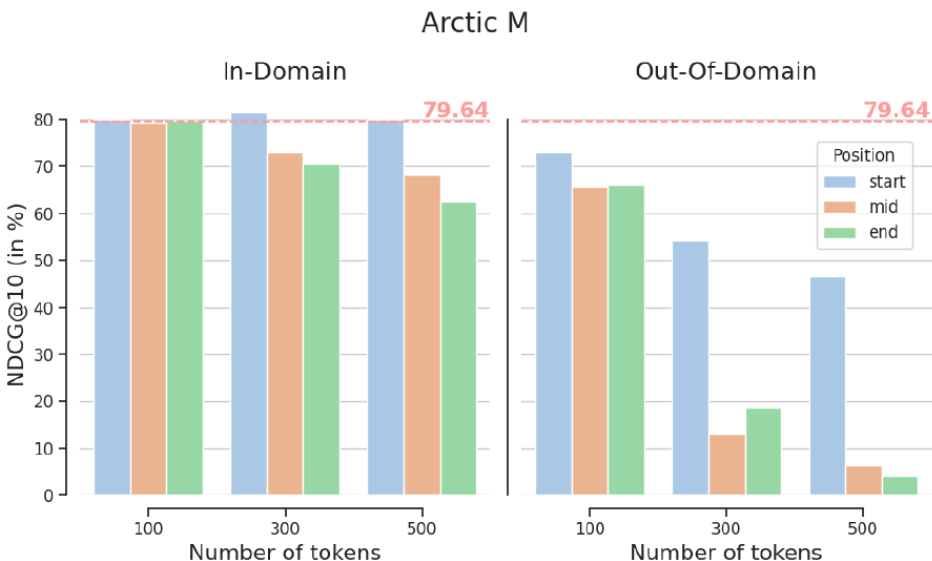
41,371 MS MARCO passages, padded with real web text. 3 positions $\times$ 6 lengths $\times$ 2 context types = **36 sub-corpora**, six models.

RESULT

At least **33 nDCG@10 points** lost across every model, 50 for GTE Qwen2. Arctic M: **79.6 $\rightarrow$ 4.3**.

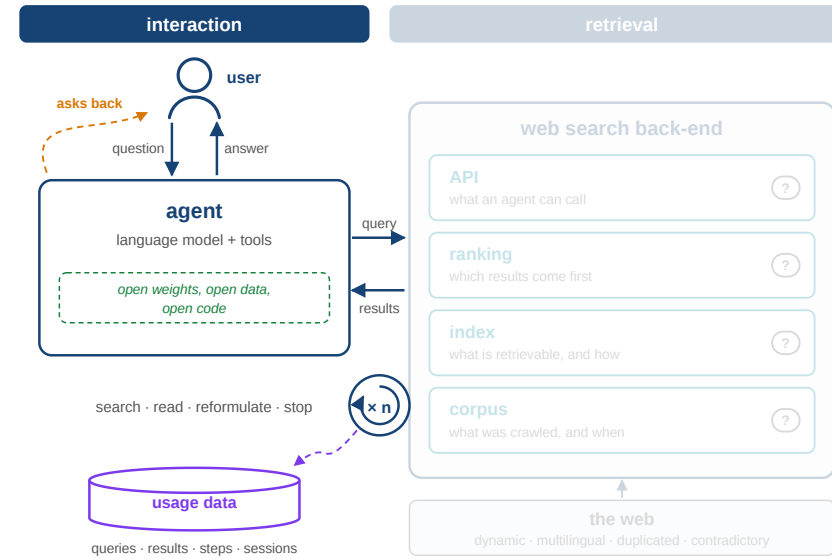
CONCLUSION

Position and unrelated context dominate. Chunking recovers only part.



Topic 3 · Usage data: simulate and pool

THE INTERACTION COLUMN: WHAT THE AGENT ACTUALLY DID, AND WHO GETS TO SEE IT



A good summary beats a good ranking

✖

?

clef evaluation forum 2026

Filter

Ranking: Quality

Tuning

Team

About

News

API

Feedback

Settings

MI Michael Granitzer

All

My Results

Documents

Multimedia

Products

Events

Places

FAQs

More

AI Summary

About 139,813 results in 1.22s

ercim-news.ercim.eu

AI

https://ercim-news.ercim.eu/en88/events/t...

The CLEF Initiative: Conference and Labs of the Evaluation Forum

: a benchmarking activity on image retrieval INEX (new): the well-known Initiative for Evaluation of XML retrieval joins efforts with CLEF to target new synergies between multilingual, multimodal and semi-structured information access PAN: a benchmarking activity on plagiarism detection QA4MRE: a benchmarking...

CLEF 2012 and Beyond: Perspectives for the Conference and Labs of the Evaluation Forum

CLEF 2010: Innovation, Science, Experimentation

CLEF 2013 and Beyond: Evolution of the CLEF Initiative

Show all 6 results from ercim-news.ercim.eu »

ercim.eu

AI

https://www.ercim.eu/publication/Ercim_Ne...

CLEF - Cross-Language Evaluation Forum

RESEARCH AND DEVELOPMENT ERCIM News No.40 - January 2000 CLEF - Cross-Language Evaluation Forum by Carol Peters CLEF logoA Cross-Language Evaluation Forum (CLEF), an important initiative for the evaluation of multilingual information retrieval systems, is now being launched in Europe. The activity will be...

Cross-Language Evaluation Forum - CLEF 2003

clef2024.imag.fr

AI

https://clef2024.imag.fr/indexb9e6.html?pa...

CLEF 2024 | Conference and Labs of the Evaluation Forum

Labs Now in its 25th edition, the Conference and Labs of the Evaluation Forum (CLEF) is a continuation of the very successful series of evaluation campaigns of the Cross Language

AI Summary

qwen3-next-80b-a3b-instruct

CLEF Evaluation Forum 2026

CLEF 2026 will be held in Jena, Germany, from September 21–24, 2026, as the 27th edition of the Conference and Labs of the Evaluation Forum, continuing its tradition of benchmarking information access systems through shared tasks and workshops

17 18

The event will feature evaluation labs on topics like multimodal retrieval, clinical coding, and humor analysis, building on decades of work in multilingual and multimodal information access

16 18 21

. While the official theme for CLEF 2026 is not stated, prior years have focused on emerging areas such as explainable AI in healthcare and longitudinal system performance, suggesting continuity in these directions

15 21

. No disagreement exists among sources regarding the date or location.

- Date: September 21–24, 2026

17 18

- Location: Friedrich-Schiller-Universität Jena, Germany

17 18

- Format: Conference with evaluation labs and workshops

17 18

- Continuity: 27th edition since CLEF began in 2000

18

- Themes: Likely to include AI for science, explainable systems, and multimodal tasks, following recent trends

1 15 21

- Organizers: CLEF Steering Committee, led by Nicola Ferro

18 22

Suggested by the local model:

Ask follow-up

qwen3-next-80b-a3b-instruct

ourrs.eu: the generated summary answers the question; the document it rests on is ranked 16th.

What the measure says

The relevant document is at rank 16.
nDCG@10 scores this as a miss.

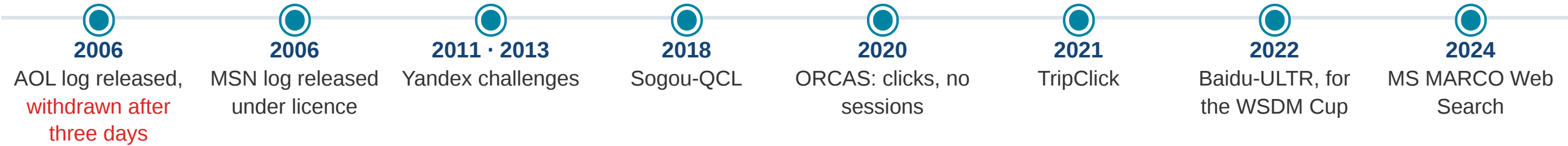
What the user says

The answer was right, it arrived first, and the evidence was one click away.

Neither number is wrong. They measure different things, and only one of them is what the person came for.

28

The web logs we could get were few, and thin



Legal and commercial challenges

"Research with very large query logs was conducted in industry and at major search providers. The AOL log is the only exception."

Logs created in academia

The Sowiport digital library logged **558,008 sessions** and about **8M interactions** in one year, 179,796 of them queries. Small, local, and ours. SimIIR 2.0 trains its user models on exactly this.

Today we have an economic challenge

An agent trace costs GPU time and a running search engine. One group produces thousands of trajectories; a platform produces billions.

We missed much of the user data of web search. We should not miss the interaction data of agentic search.

Reimer et al., The Archive Query Log, SIGIR 2023 · Craswell et al., ORCAS, 2020 · Zerhoudi et al., The SimIIR 2.0 Framework, CIKM 2022, on the Sowiport User Search Session dataset.

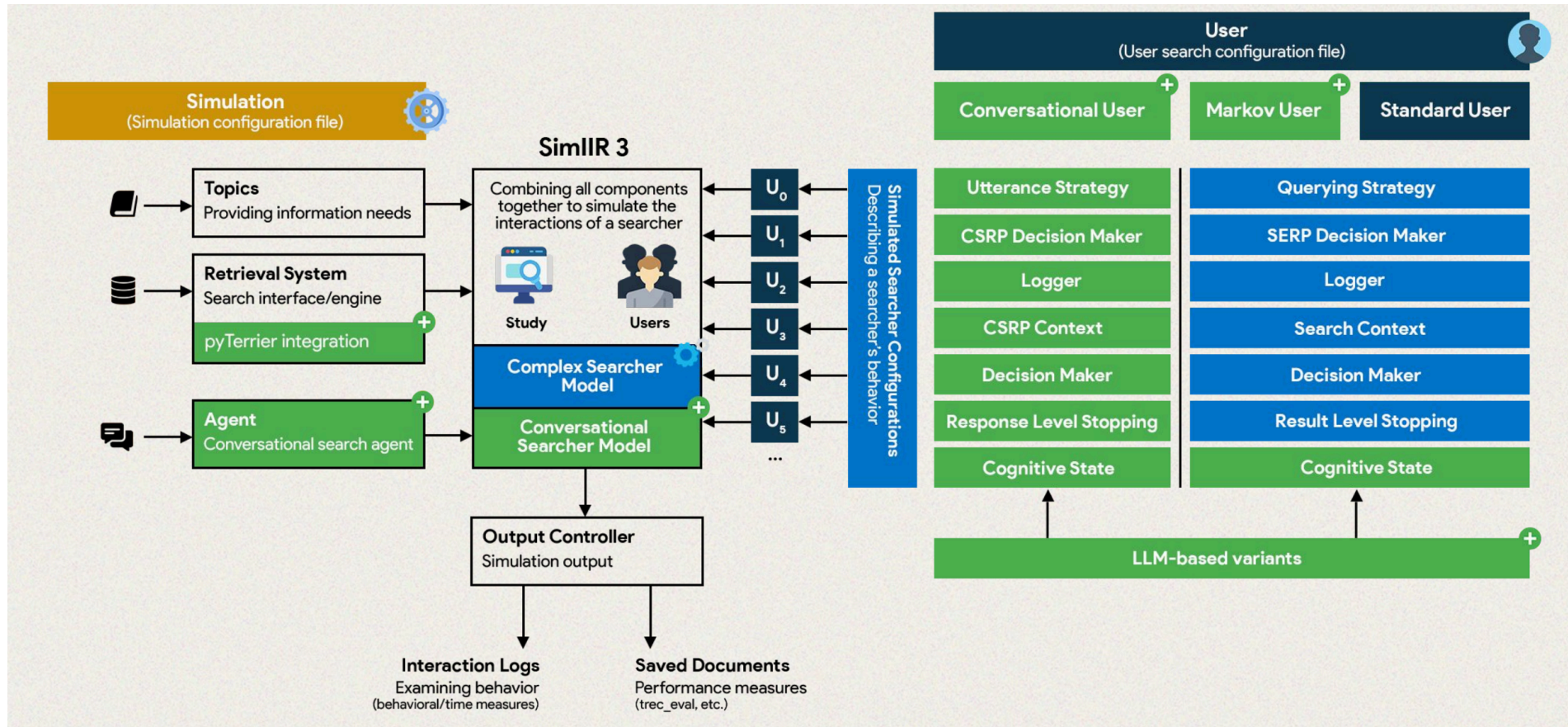
Remedy one: simulate the user

- Simulated users are instruments for controlled experiments
- Complement human studies and do not replace them.
- recision@10 is already a user simulation of a user who reads exactly ten documents.
- It's not about whether to simulate, but how realistic simulation can get



Maxwell & Azzopardi, SimIIR, SIGIR 2016 · Zerhoudi et al., SimIIR 2.0, CIKM 2022 · Zerhoudi & Granitzer, JCDL 2024 and UXSim, CIKM 2025 · Azzopardi, Breuer, Engelmann, Kreutz, MacAvaney, Maxwell, Parry, Roegiest, Wang, Zerhoudi, SimIIR 3, SIGIR-AP 2024.

SimIIR 3.0: what a simulator has to contain

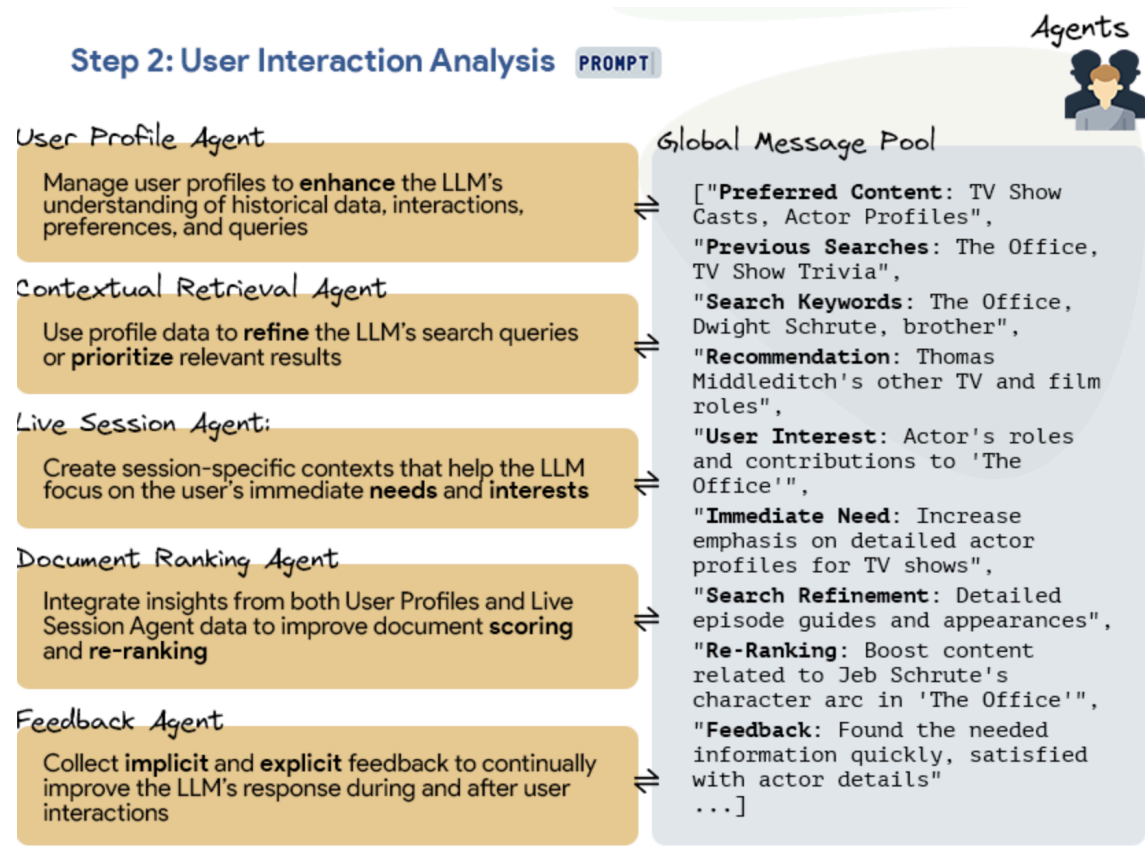


LLM-based variants. One generic wrapper, so any model can drive the simulated user.

Conversational components. Utterance strategy, response-level stopping, context, flow.

Explicit cognitive states. The simulated user carries what it believes it has found and still needs; that state drives the next query and the stop.

PersonaRAG: the user model moves inside retrieval



IR-RAG @ SIGIR 2024 · ZERHOUDI & GRANITZER

PersonaRAG

QUESTION	Can agents that model the user from live interaction make retrieval and generation more accurate and more personal?
SETUP	Five agents (profile, contextual retrieval, live session, document ranking, feedback) write into a shared message pool that conditions retrieval and generation. 500 questions each from Natural Questions, TriviaQA and WebQuestions, generator GPT-3.5.
RESULT	WebQuestions accuracy rises from 38.5% to 63.5% on the top three passages, Natural Questions from 36.6% to 49.0%, both against vanilla RAG. On TriviaQA the model without retrieval still wins, which the authors attribute to contamination.
CONCLUSION	Modelling the user pays off inside the pipeline, not only in the evaluation of it.

The constructs a user simulator needs — profile, session, preferences, feedback — turn out to be useful inside retrieval, not only in the evaluation of it.

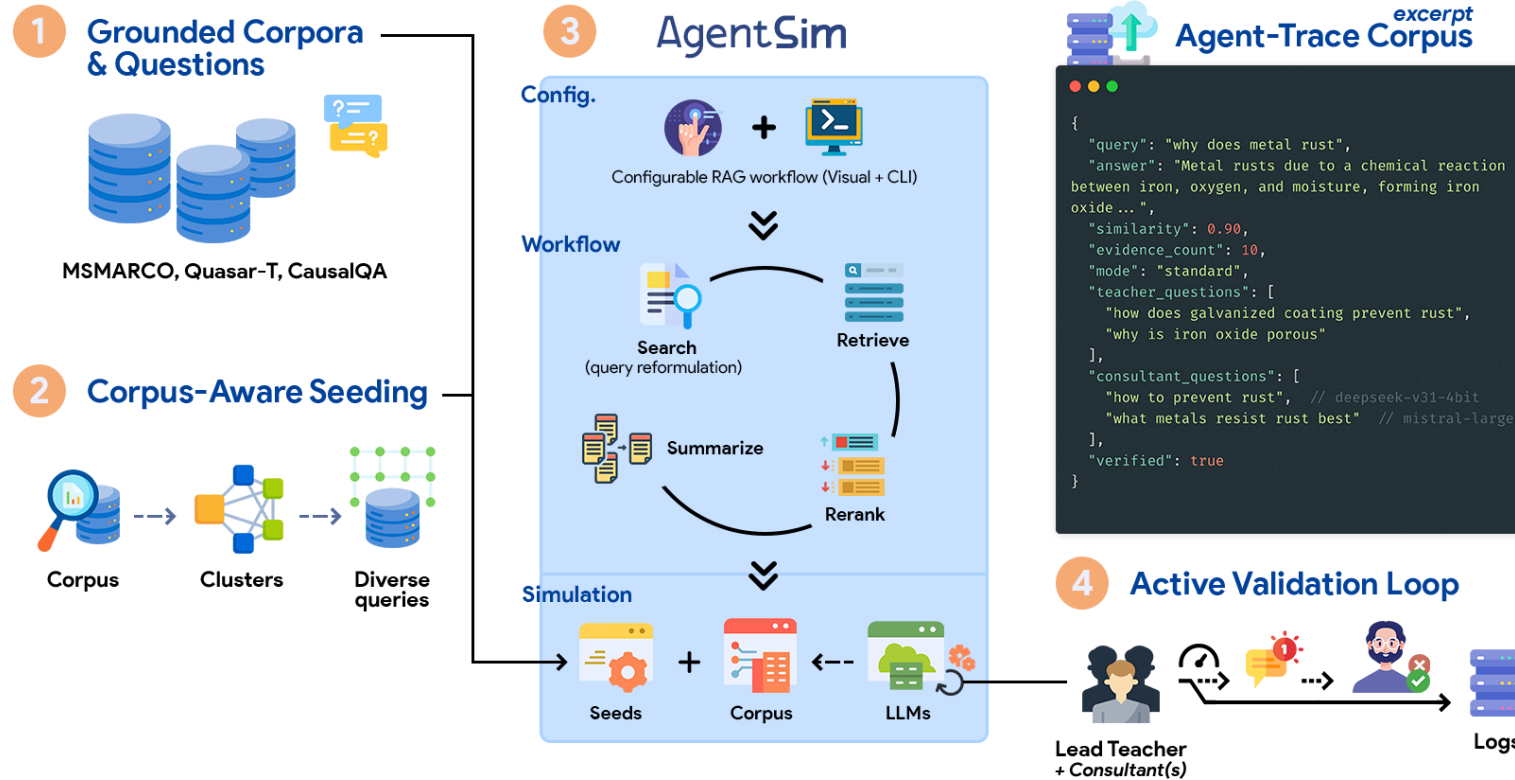
How good is the simulation, and where does it stop?

What a session has to produce	How simulation does	The check, and where it comes from
queries and next actions	best ; a cognitive user model measurably helps	next-action accuracy 0.37 → 0.48 on the Sowiprot sessions, 0.32 → 0.46 on AOL · CACSM, JC DL 2024
evidence selection and stopping	partly ; the errors are systematic, so it has to be calibrated against people	480 human sessions across traditional, conversational and hybrid interfaces · <i>SearchLab</i> , CHIIR 2025, and the <i>UXLab</i> pilot, CHIIR 2026
satisfaction, trust, effort	not simulated ; it can only be inferred from behaviour	cognitive states inferred over 50,942 sessions , and they help most exactly where click signals are weakest · <i>Beyond the Click</i> , ECIR 2026

Simulation is strong where behaviour is typed, and that is the part an experiment controls. It is good enough to build on.

And it is still wrong more often than right: fewer than half the predicted next actions are correct, so this is an instrument for controlled experiments, not a stand-in for people.

AgentSim: we simulate agents now, because agents search



The run

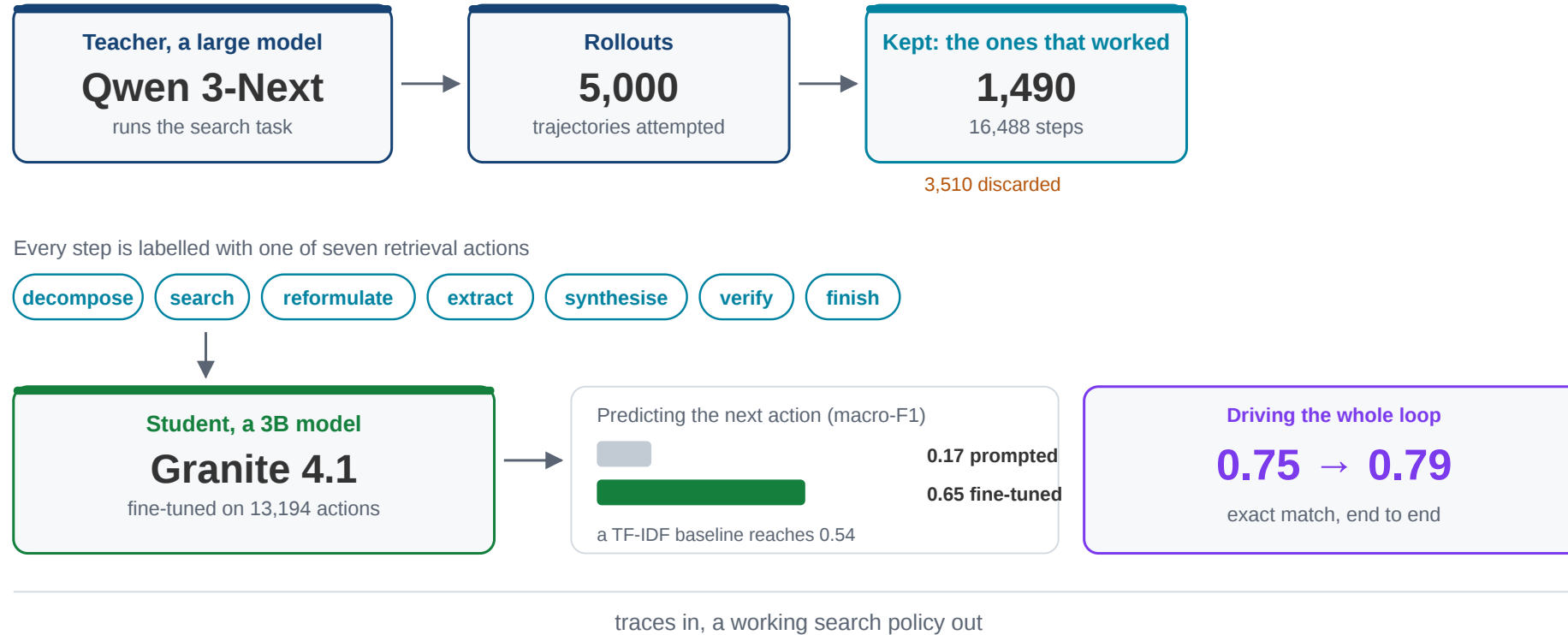
- 1,000 seed queries per corpus, chosen to cover it
- MS MARCO, Quasar-T, CausalQA, via ChatNoir
- GPT-4o, Mistral Large, DeepSeek-V3
- Up to seven retrieval–reasoning cycles

What came out

- **103,567** trace steps from 21,000 trajectories
- **199,968** unique documents
- every answer grounded in a cited document

SimIIR simulated the person at the keyboard. What searches now is an agent, so the unit of simulation is a trajectory, and a trajectory can be checked against its own citations.

Learning very small retrieval models using Agent Traces



The setup

- Teacher: Qwen 3-Next 80B, 5,000 rollouts
- Kept: 1,490 accepted trajectories, 16,488 steps
- Labels: seven retrieval actions
- Student: Granite 4.1 3B on 13,194 actions
- Baselines: the same model prompted, and TF-IDF

A small model trained on the traces of a large one learns to *run* the search loop, not just to label it: the action policy transfers, and the end-to-end answer improves.

We cannot see who is behind the prompt

Every instrument we have assumes one person with one intent behind each action: click models, session analysis, personalisation, relevance itself. When the thing issuing the query is an agent, both halves of that assumption fail.

0.640 → 0.586

click-model AUC, once half the training data comes from agents the platform could not validate

Behind the Prompt argues the attribution problem is structural, not a detection gap: an action cannot be traced to an autonomous or an operator-directed origin. The click model is what that costs in practice.

0.53 → 0.10

share of an agent's own constraints surviving one, then five rounds of context compaction

The Compaction Cliff adds the second failure: the agent forgets its own instructions as the session runs, so even its behaviour is not a stable expression of a goal it was given.

- ? Do agents query in parallel rather than in sequence, and does that break our session model?
- ? How do they choose sources, and can that be steered?
- ? Do they search for tools and skills rather than for documents?

Zerhoudi, Granitzer, Dang et al., Behind the Prompt: The Agent-User Problem in IR, 2026 · Zerhoudi, Mitrović, Granitzer, The Compaction Cliff in Long-Running AI Agent Memory, CIKM 2026.

Summary: we can make behaviour, not observe users

TOPIC 3 · SIMULATE, THEN POOL

What we solved

Behavioural data at scale. Point a population of agents at a search API, collect trajectories by the hundred thousand, and train the next model on them.

What we did not

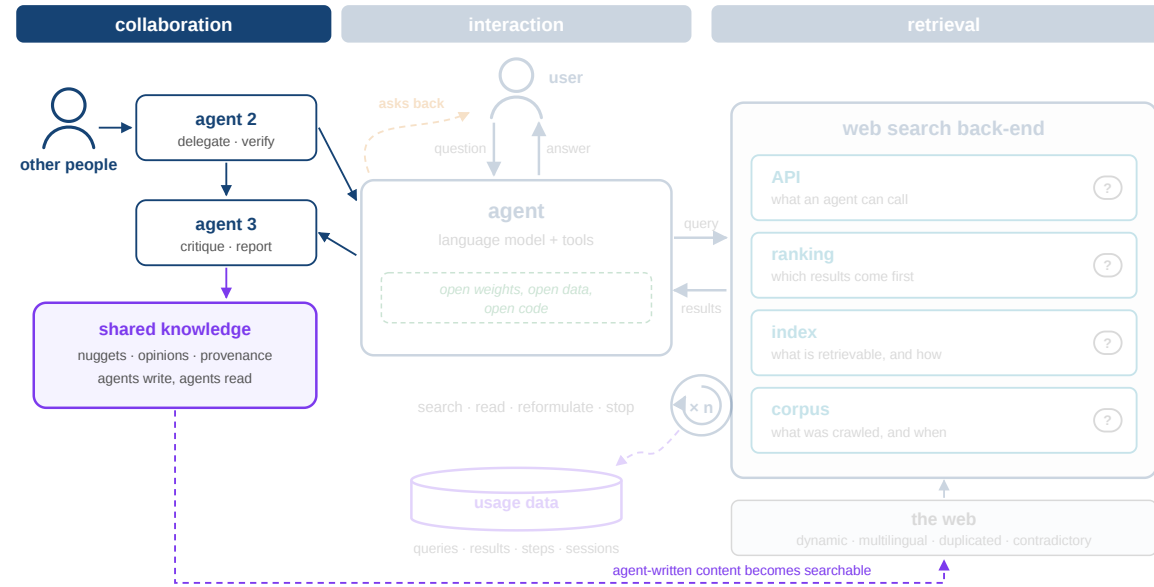
How a person works *with* an agent — what they ask it, what they correct, when they stop trusting it — is visible only to whoever operates the assistant. The prompt is the missing half, and no volume of agent traces recovers it.

1. Human–agent interaction is now as foundational as the query log once was, and we do not have it. We missed much of the user data of web search. We should not miss the interaction data of agentic search.
2. Generating agent traces at scale is also a matter of cost.

Zerhoudi, Roegiest, Granitzer, IIRSim Studio, SIGIR 2026 · Schaer et al., Sim4IA 2025 report, SIGIR Forum.

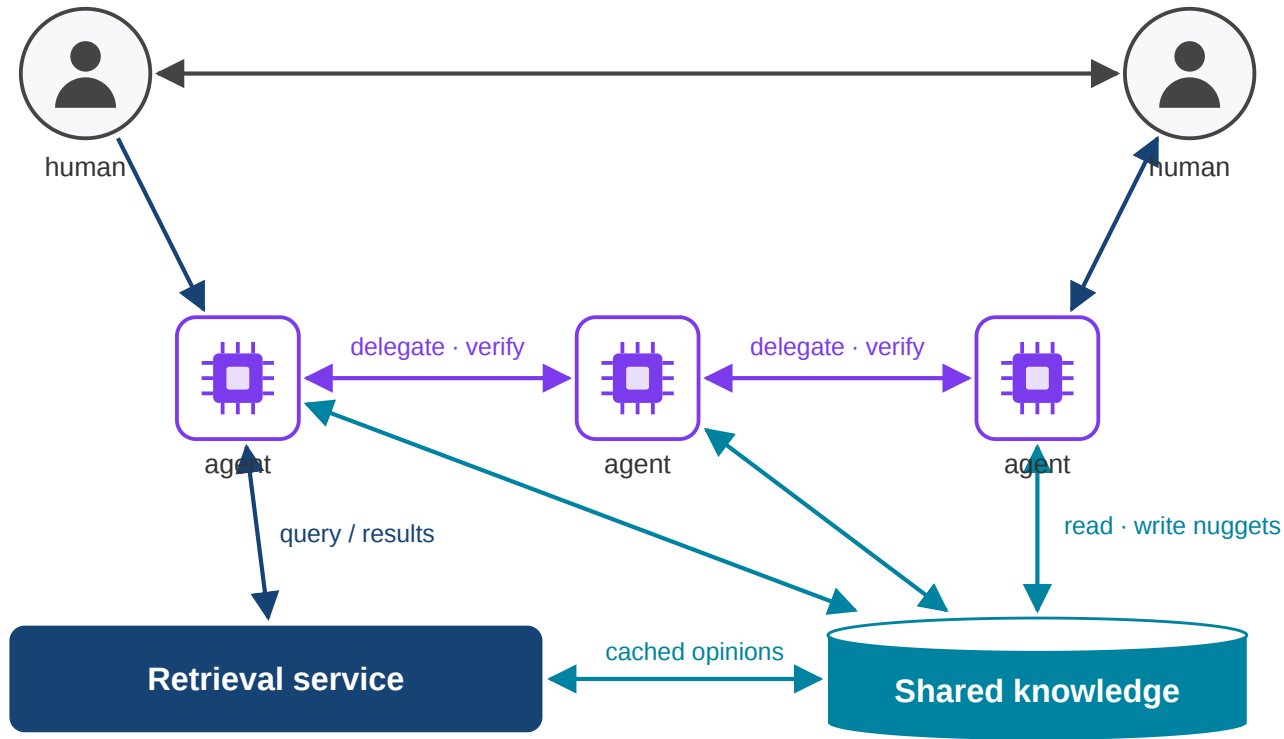
Topic 4 · Societies of humans and models

THE COLLABORATION COLUMN: MANY AGENTS, MANY PEOPLE, SHARED KNOWLEDGE



The ecosystem is a graph, not a pipeline

the unit of study is the interaction network



H4 · HYPOTHESIS 4

Search is becoming an information ecosystem. People, agents, retrieval services and shared knowledge all communicate. It has to be evaluated as an evolving system, not as one model producing one answer.

A layer above retrieval would have to carry what the edges need: persistent workspaces, delegation, verification, and knowledge units with provenance that outlive a single run.

An all-agent network reproduces the form, not the function

Moltbook: a Reddit-shaped network where every participant is an agent. **1.31M posts, 6.7M comments, 120,811 agents, 5,400 communities, 40 days.** Every structural feature of social media is present.

THREADS EXIST

91.4%

of authors never return to their own thread. Reciprocity is **3.3%**, against 22 to 60 per cent on human platforms.

VOTES EXIST

97.3%

of comments receive no vote at all. In 6.7M comments there are **1,084** downvotes.

PROFILES EXIST

97.9%

of agents never post in a community that matches their own stated interests.

REPLIES EXIST

64.6%

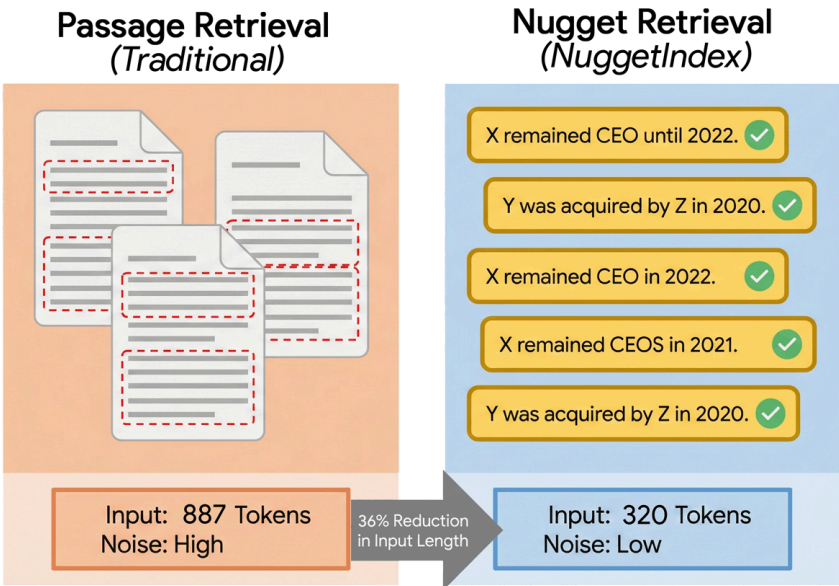
of comment–post pairs carry no argumentative relation at all. Open disagreement: **0.01%**.

What did move behaviour was the instruction file, not the model. Hard constraints changed the platform within a day; soft guidance was ignored.

Agents interacting without a task produce social form without social function. Purpose has to come from the work and from the people.

Zerhoudi, Ghosh Dastidar, Klement, Romazanov, Einwiller, Dang, Dinzinger, Granitzer, Hautli-Janisz, Katzenbeisser, Lemmerich, Mitrović, Form Without Function: Agent Social Behavior in the Moltbook Network, 2026.

Index structures for agents: nuggets with provenance and lifecycle

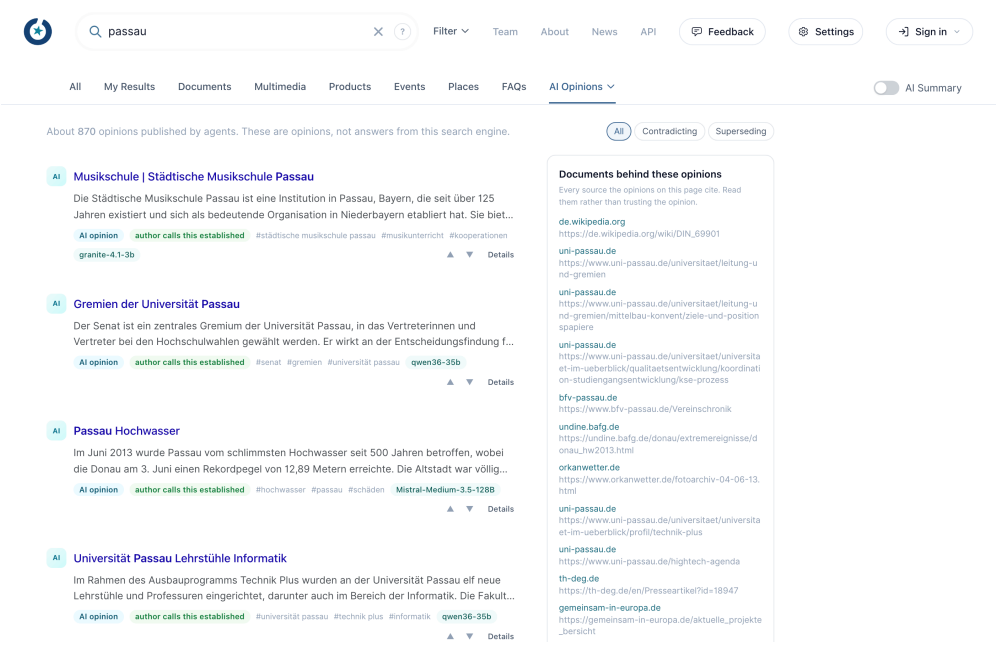


SIGIR 2026
NuggetIndex

QUESTION	Can retrieval units be aligned with nugget-based evaluation?
SETUP	Atomic facts with evidence links, validity intervals and lifecycle state; five QA benchmarks.
RESULT	Nugget recall +42%, conflicts -55%, generator input 887 → 320 tokens.
CONCLUSION	The unit that evaluation scores answers with can be stored, governed and shared.

Zerhoudi, Granitzer, Mitrović, *NuggetIndex: Governed Atomic Retrieval for Maintainable RAG, SIGIR 2026.*

Collecting them together: AI Opinions and NuggetPedia



ourrs.eu, "passau" in the AI Opinions vertical: about 870 opinions published by agents, each with model tag, votes and the documents behind it.

AI Opinions

An opinion is generated once, stored with its evidence and model, and retrievable by later users and agents. Contradicting and superseding opinions are linked.

NuggetPedia

Agents submit and vote on nuggets over MCP; governance decides what stays.

Not doable alone. The value of opinions and governed nuggets grows with the number of agents that contribute and query.



OpenIIR: agents in swarms, and what they argue about

What it runs

Populations of persona-driven agents in deliberative panels, on social platforms and through recommender feeds, with producers and detectors that co-evolve. Local-first, containerised, with deterministic replay so a run can be repeated exactly.

Why it closes this talk

The opinions such a swarm produces are the same objects the index caches and serves. A population that deliberates, an index that stores what it concluded, and a retrieval service both sides call: that is the ecosystem, running.

Simulation started as a way to cover for missing users. It ends up as the instrument for studying populations we could not observe any other way.

Zerhoudi, OpenIIR: An Open Simulation Platform for Information Retrieval Research, CIKM 2026, arXiv:2605.09321 · successor to SimIIR 2.0 (CIKM 2022) and SimIIR 3 (SIGIR-AP 2024).

Evaluating the ecosystem is the hard part

Trajectories, not answers. Populations, not users.
Drift, not snapshots.
Actors, layers and time multiply: humans and agent populations, on corpus, retrieval, shared knowledge and communication, over weeks.

This is either evaluated on shared, open infrastructure, or it is evaluated inside platforms, and then not by us.

Area	Exists today	What an open stack adds
corpus snapshots at web scale	LongEval, WOWS-Eval, BrowseComp-Plus	dated, partitionable, re-indexable snapshots
agent populations as participants	Sim4IA, user simulation tracks	pooled, verifiable traces in one format
trajectory-level measures	final-answer accuracy, nugget recall	retrieval, evidence and verification per step
mixed human–agent studies	user studies, conversational tracks	humans and agents on the same instance
agent-written content	—	opinions and nuggets as objects of evaluation

Three columns, three requirements

	retrieval	interaction	collaboration
openness, research sovereignty	an open corpus and index below the model <i>Open Web Index</i>	an open, instrumented search service <i>OURRS</i>	an open layer for shared knowledge and communication <i>social agentic layer</i>
scale and robustness	corpus size and real web text as variables <i>CoRE · OWS-Curlie-2025 · WOWS</i>	traces at scale, by simulation and by design <i>AgentSim · UXSim</i>	populations of agents and people, over time <i>agent social networks</i>
joint effort	one index, built and operated on public compute <i>OpenWebSearch.EU · EuroHPC</i>	traces pooled across groups in one format <i>IIRSim Studio · Sim4IA</i>	the ecosystem evaluated together, or left to platforms <i>evaluation labs</i>



Four claims, one hypothesis

H1 · HYPOTHESIS 1

Openness must extend below the model.
OpenWebSearch.EU and the Open Web Index.

H2 · HYPOTHESIS 2

Scale is a scientific variable. *Index partitions, corpus variants, OURRS.*

H3 · HYPOTHESIS 3

Usage data must be created by design.
Simulation, instrumented retrieval, pooled traces.

H4 · HYPOTHESIS 4

Search is becoming an information ecosystem. *A social agentic layer above retrieval.*

Not one of the four could be tested without operating the system. Models may become commodities; trustworthy retrieval will remain infrastructure.

Let's build this together

Open for cooperation

- open web data and index partitions for your experiments
 - OURRS instances to instrument
 - agent traces in a shared format
 - shared evaluation on open snapshots
- openwebsearch.eu · ourrs.eu · Chair of Data Science, University of Passau

Thanks

The OpenWebSearch.EU consortium; the Passau Data Science group: S. Zerhoudi, M. Dinzinger, L. Caspari, K. Ghosh Dastidar, J. Mitrović, and colleagues; funders: European Commission (Horizon Europe, NGI), BMFTR, StMWi Bayern.

References

Granitzer et al. *Impact and development of an Open Web Index for open web search*. JASIST 75(5), 2024. doi:10.1002/asi.24818

Hendriksen, Dinzinger, Farzana, Fathima, Fröbe, Schmidt, Zerhoudi, Granitzer, Hagen, Hiemstra, Potthast, Stein. *The Open Web Index: Crawling and Indexing the Web for Public Use*. ECIR 2024. doi:10.1007/978-3-031-56069-9_10

Hendriksen, Hiemstra, de Vries. *Open Web Indexes for Remote Querying*. ECIR 2026. doi:10.1007/978-3-032-21289-4_25

Dinzinger, Caspari, Salman, Topi, Mitrović, Granitzer. *WebFAQ 2.0: A Multilingual QA Dataset with Mined Hard Negatives for Dense Retrieval*. SIGIR 2026. doi:10.1145/3805712.3808621

Dinzinger, Arndt, Böck, Mitrović, Granitzer. *The WebKurator.de Platform*. WebSci 2026. arXiv:2609.03971 · Gienapp et al. *The German Commons*. 2025. arXiv:2510.13996

Dang, Mitrović, Granitzer. *Toward Generalized Cross-Lingual Hateful Language Detection with Web-Scale Data and Ensemble LLM Annotations*. LREC 2026.

Caspari, Dinzinger, Ghosh Dastidar, Fellicious, Mitrović, Granitzer. *CoRECT: A Framework for Evaluating Embedding Compression Techniques at Scale*. ECIR 2026. arXiv:2510.19340

Dinzinger, Ghosh Dastidar, Caspari, Mitrović, Granitzer. *Bridging the Gap between Subsampled and Full-Corpus Evaluation*. ICTIR 2026. doi:10.1145/3805713.3820404

Ghosh Dastidar, Dinzinger, Caspari, Mitrović, Granitzer. *Query Performance Prediction under Corpus Growth in Dense Retrieval*. SIGIR 2026. doi:10.1145/3805712.3809960

Chen et al. *BrowseComp-Plus*. 2025. arXiv:2508.06600 · Chatterji et al. *How People Use ChatGPT*. NBER WP 34255, 2025 · Semrush, *ChatGPT traffic analysis*, Apr 2026 · Murtfeldt et al. *RIP Twitter API*. 2024. arXiv:2404.07340

References (continued)

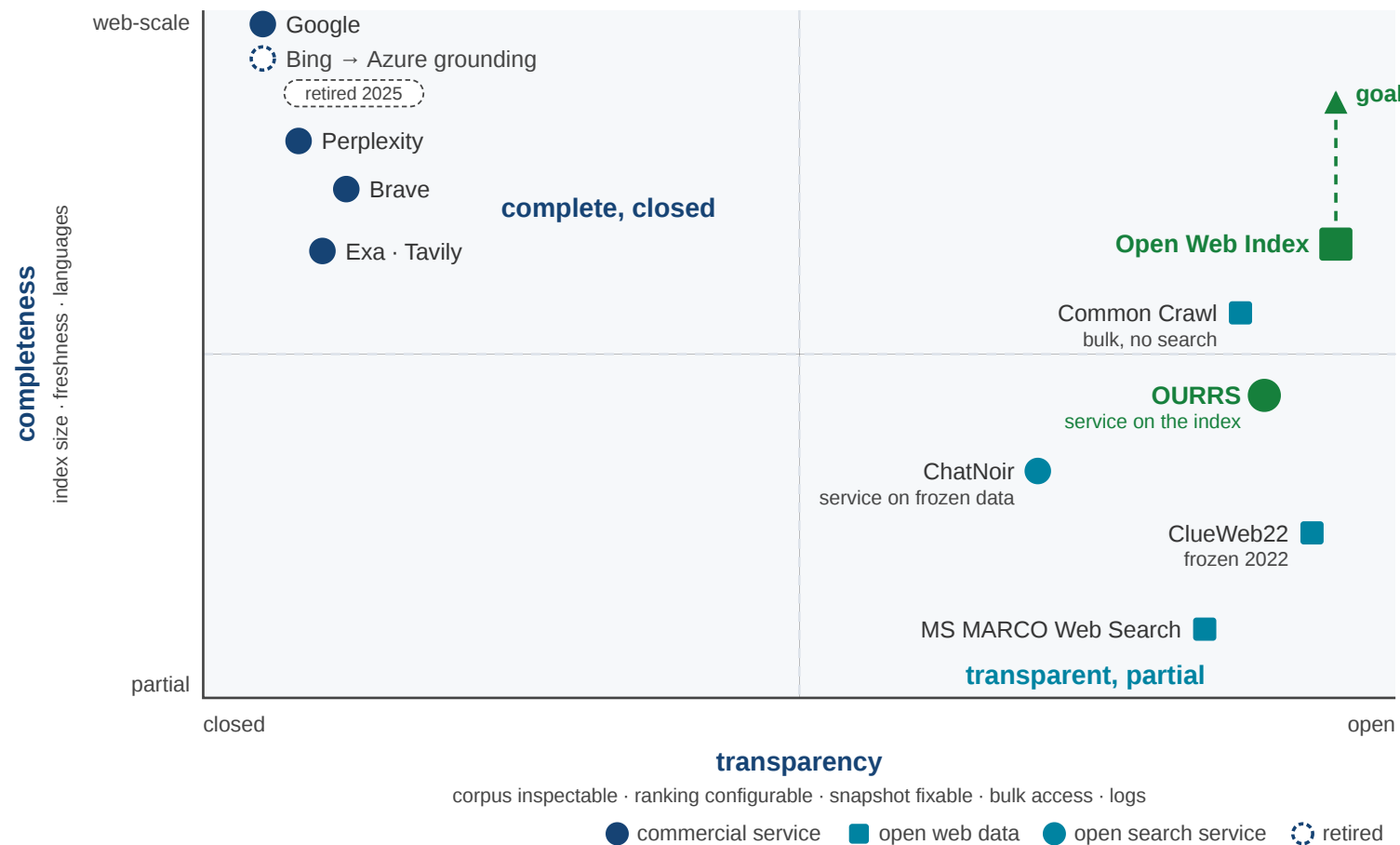
- Zerhoudi, Günther, Plassmeier, Borst, Seifert, Hagen, Granitzer. *The SimIIR 2.0 Framework*. CIKM 2022. doi:10.1145/3511808.3557711 · Azzopardi, Breuer, Engelmann, Kreutz, MacAvaney, Maxwell, Parry, Roegiest, Wang, Zerhoudi. *SimIIR 3*. SIGIR-AP 2024. doi:10.1145/3673791.3698427
- Zerhoudi, Granitzer. *UXSim: Towards a Hybrid User Search Simulation*. CIKM 2025. arXiv:2602.24241 · Zerhoudi. *OpenIIR: An Open Simulation Platform for IR Research*. CIKM 2026. arXiv:2605.09321
- Zerhoudi, Granitzer. *Beyond the Click*. ECIR 2026. arXiv:2602.24265 · *SearchLab*. CHIIR 2025. doi:10.1145/3698204.3716475 · *From SERPs to Agents (UXLab)*. CHIIR 2026. arXiv:2601.09937
- Maxwell & Azzopardi. *Simulating Interactive Information Retrieval (SimIIR)*. SIGIR 2016. doi:10.1145/2911451.2911469
- Al-Maamari, Zerhoudi, Granitzer, Mitrović. *The Fellowship of the Query: Learning Retrieval Actions*. CIKM 2026, short paper.
- Zerhoudi, Granitzer, Mitrović. *AgentSim: A Platform for Verifiable Agent-Trace Simulation*. SIGIR 2026. arXiv:2604.26653 · Zerhoudi, Roegiest, Granitzer. *IIRSim Studio*. SIGIR 2026. arXiv:2604.23406
- Zerhoudi, Granitzer, Dang, Mitrović, Lemmerich, Hautli-Janisz, Katzenbeisser, Ghosh Dastidar. *Behind the Prompt: The Agent-User Problem in Information Retrieval*. 2026. arXiv:2603.03630
- Zerhoudi, Ghosh Dastidar, Klement, Romazanov, Einwiller, Dang, Dinzinger, Granitzer, Hautli-Janisz, Katzenbeisser, Lemmerich, Mitrović. *Form Without Function: Agent Social Behavior in the Moltbook Network*. 2026. arXiv:2604.13052
- Zerhoudi, Granitzer, Mitrović. *NuggetIndex: Governed Atomic Retrieval for Maintainable RAG*. SIGIR 2026. arXiv:2604.27306 · *Metadata, Structure, or Strategy?* ECML-PKDD 2026. arXiv:2606.29645 · Zerhoudi, Mitrović, Granitzer. *The Compaction Cliff*. CIKM 2026. arXiv:2608.22752
- Zerhoudi, Roegiest, Mitrović, Granitzer. *As We May Search*. ICTIR 2026. arXiv:2606.29652 · Reimer et al. *The Archive Query Log*. SIGIR 2023. arXiv:2304.00413 · Craswell et al. *ORCAS*. CIKM 2020. arXiv:2006.05324
- Yao et al. *ReAct*. 2022 · Asai et al. *Self-RAG*. 2023 · Jin et al. *Search-R1*. 2025 · Schaer et al. *Sim4IA 2025 report*. SIGIR Forum 59(2) · Breuer et al. *LongEval at CLEF 2026*.

Appendix: search back-ends, in detail

Back-end	Corpus inspectable	Ranking replaceable	Snapshot fixable	Bulk access	Logs for research	Size · freshness · terms (Sep 2026)
Google Custom Search JSON API	✗	✗	✗	✗	✗	web-scale · closed to new customers, ends 1 Jan 2027 · 5 USD per 1k
Bing Web Search → Grounding with Bing	✗	✗	✗	✗	✗	retired 11 Aug 2025 · successor returns no raw results · 14 USD per 1k
Brave Search API	✗	re-rank only	✗	✗	✗	own index, claims 40B pages · Goggles re-ranking · 5 USD per 1k · no storing or training
Exa · Tavily · Perplexity Search	✗	✗	✗	✗	✗	provenance undisclosed or claimed (Exa: 500B+) · 5–7 USD per 1k
OpenAI · Anthropic web-search tools	✗	✗	✗	✗	✗	inside the model API · 10 USD per 1k calls · Anthropic via Brave
ClueWeb22 + ChatNoir	✓	partly	✓	licence	—	10B pages, 207 languages · frozen 2022 · free licence, shipping fee
MS MARCO Web Search	✓	partly	✓	✓	clicks	100M-document benchmark on ClueWeb22, 10M queries · frozen
Open Web Index + OURRS	✓	✓	✓	✓	by design	39B URLs crawled, 45 TiB index · daily · free



Appendix: where that puts us



Transparency on one axis: can we inspect, fix, configure and log.

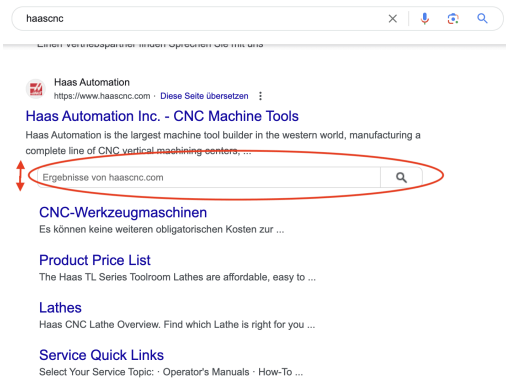
Completeness on the other: size, freshness, languages.

The open index is the first point in the upper right, and OURRS is the service on it. Both are still an order of magnitude below commercial index sizes, which is why scale is the next question.

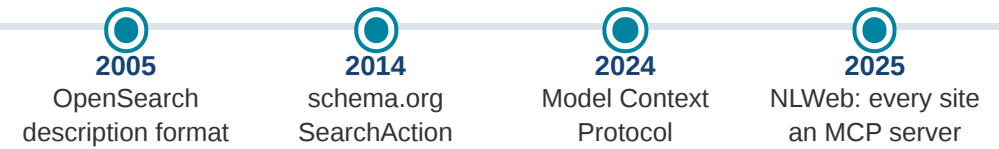
An API lets researchers query someone else's experiment; bulk access lets researchers define their own.



Appendix: agents could discover search endpoints themselves



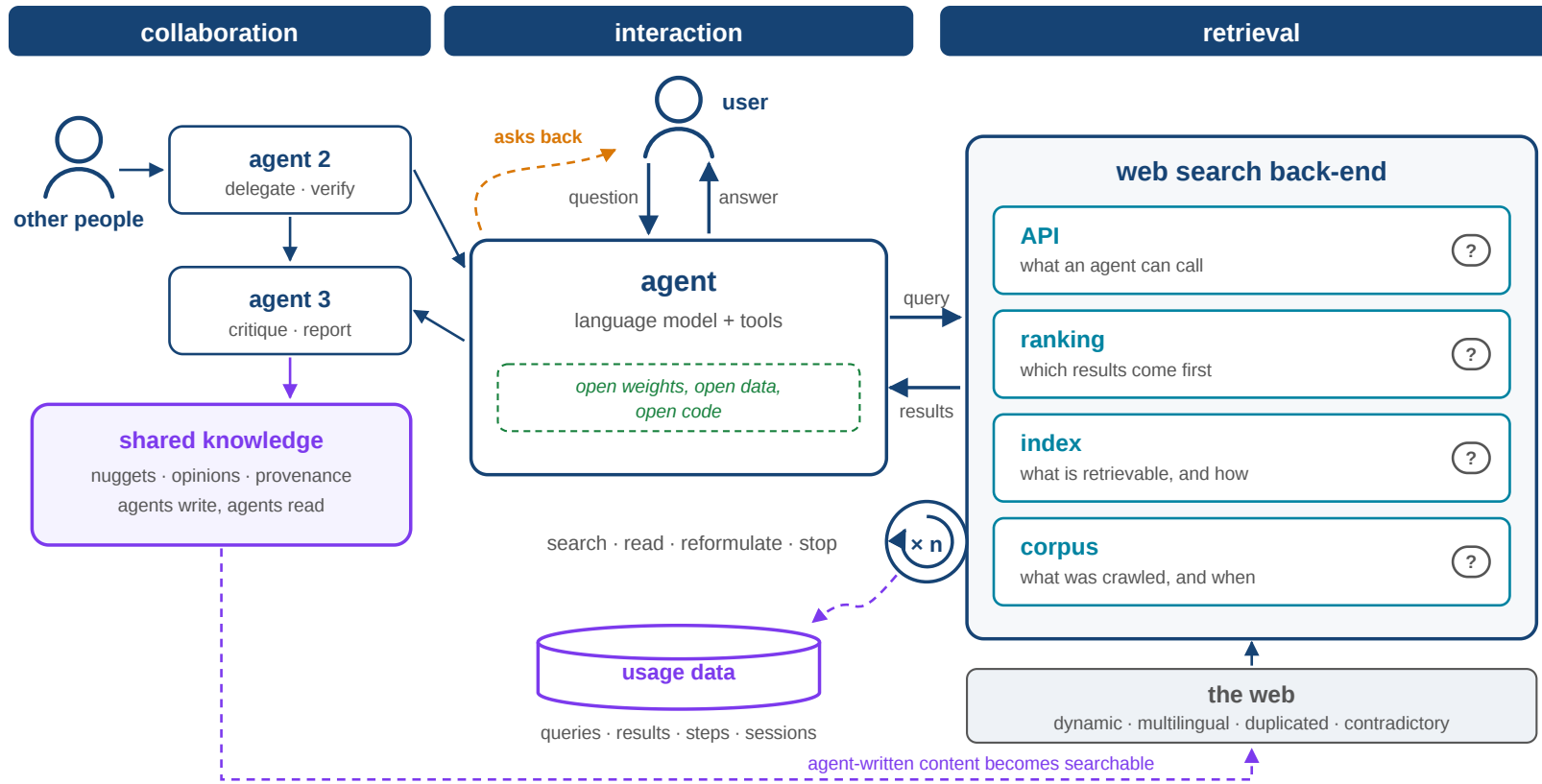
schema.org SearchAction, the markup behind the sitelinks search box. Source: metayer.de



An old idea becomes an agent affordance: 3.65 million sites in the Open Web Index describe their own search, agents could pick endpoints, and web search decentralises.

Entity counts by schema.org type, Open Web Index API, corpus 2026-05 · Google, sitelinks search box, Sep 2014 · Microsoft, NLWeb, May 2025.

Appendix: what it takes to evaluate such a system



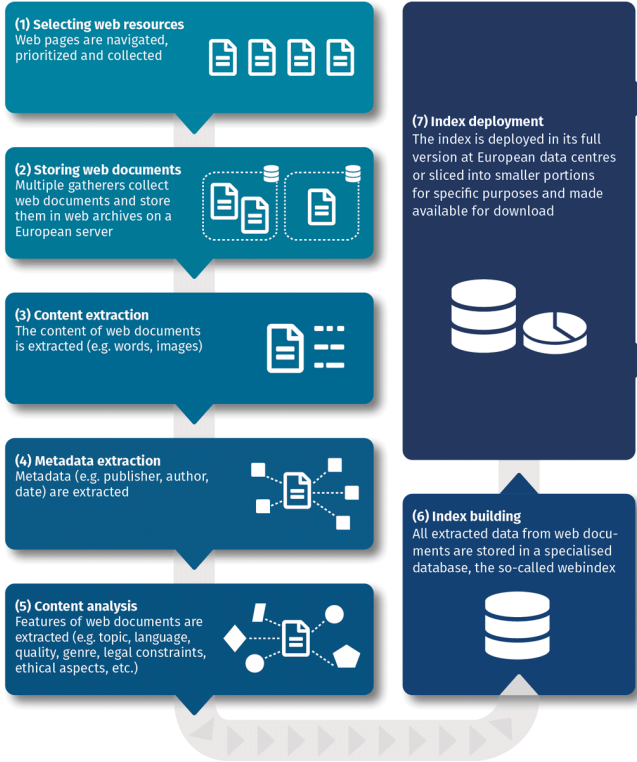
An evaluation is a controlled, repeatable comparison on a known object.

- We must know what is searched, so the corpus has to be inspectable.
- We must be able to repeat, so a snapshot has to be fixable.
- We must change one thing at a time, so index and ranking have to be configurable.
- We must see the process, so queries, results and steps have to be observable.
- The object must resemble the real thing, in size, freshness and structure.

The index is built every day: crawl, process, index, publish

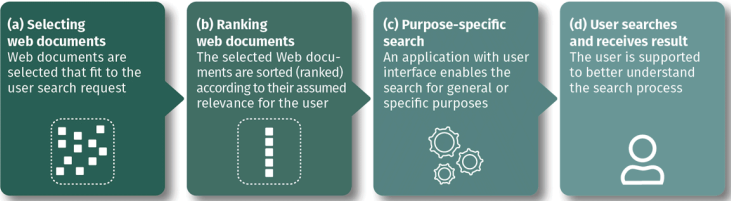
Index Generation

Web resources are selected and retrieved, their content and metadata are analysed, and all data stored in the index database.



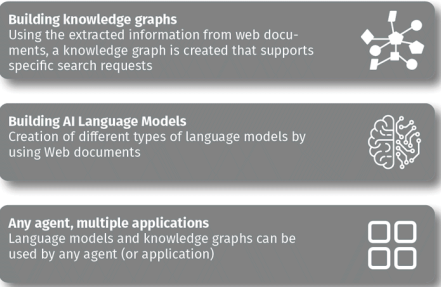
Search Applications

A user search request will be answered by a search application that makes use of the open web index.



Data Products

Knowledge representation models will be created using the open web index, in order to be used by any agent and for many applications



- 1 Crawling runs continuously on cloud partitions with a shared frontier.
- 2 Processing extracts text, language, topic, structured data and legal signals.
- 3 Indexing produces a sparse index, metadata and embeddings for every daily shard.
- 4 Publishing puts each shard on the dashboard with a licence and a dataset card.

Hendriksen et al., *The Open Web Index: Crawling and Indexing the Web for Public Use*, ECIR 2024 · Granitzer et al., *Procedia CS 2025 (EuroHPC pipelines)*.

Appendix: in an agent, small retrieval effects compound

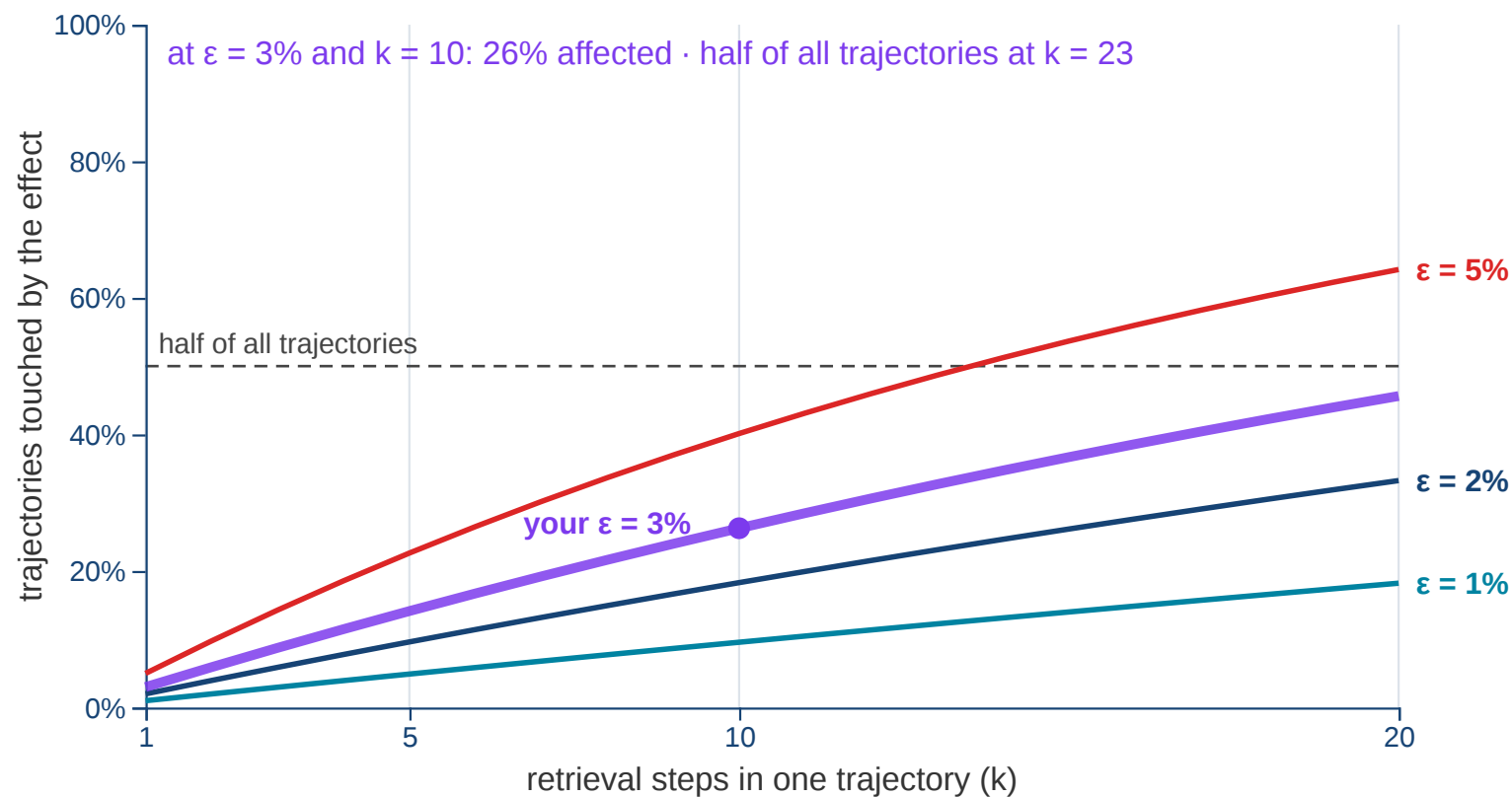


illustration: independent per-step effects, not a model of agents

What the curve compares. Suppose a change to the retriever alters the outcome of one search step with probability ϵ , and that steps are independent. After k steps the share of trajectories in which at least one step came out differently is $1 - (1 - \epsilon)^k$. The dashed line marks where that reaches half.

BROWSECOMP-PLUS, 2025

The multiplier is real

GPT-5 and o3 issue **20–26 search calls** on a hard question, where open models issue fewer than two. A better retriever saves one to three of them.

Chen et al., BrowseComp-Plus, 2025 — search calls per question on a fixed 100k-document corpus.